

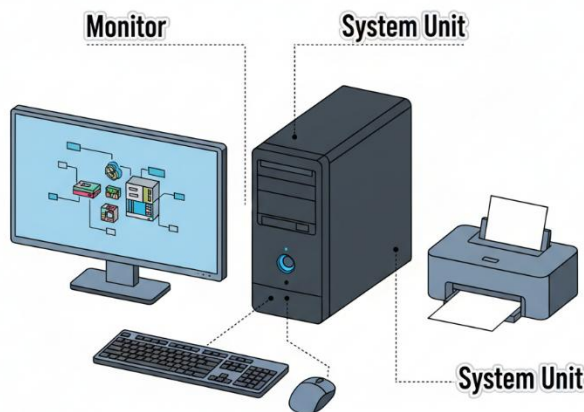
JSS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

FIRST TERM

WEEK 1: INTRODUCTION TO COMPUTER HARDWARE AND GSM REPAIRS

Introduction

In today's digital world, computers and mobile phones are among the most essential tools for communication, education, business, and entertainment. As technology advances, the need for people who can understand, maintain, and repair these devices becomes more important. This topic introduces students to computer hardware and GSM (Global System for Mobile communication) repairs—two closely related areas that provide numerous career and business opportunities.



COMPUTER HARDWARE COMPONENTS

Physical parts you can see and touch.

Meaning of Computer Hardware

Computer hardware refers to the **physical components** of a computer system that you can see and touch. These parts work together to allow the computer to perform different functions such as processing data, storing information, and displaying results.

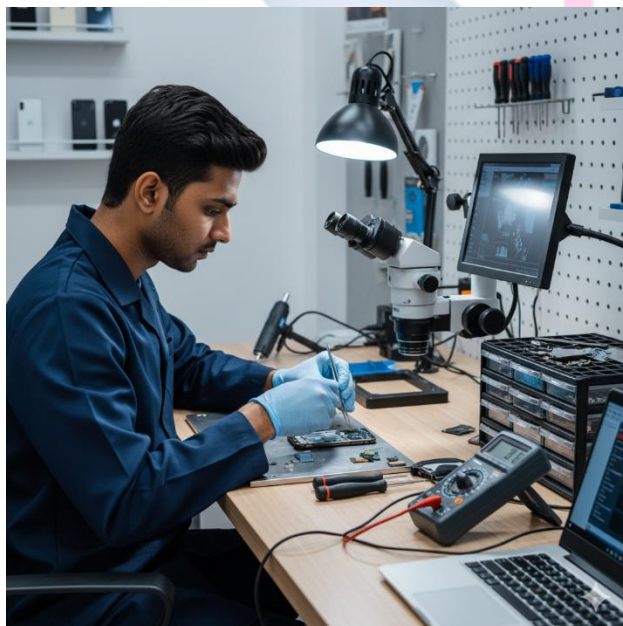
Examples of computer hardware include:

- **Monitor:** Displays information and images.
- **Keyboard:** Used to input data.
- **Mouse:** Helps in selecting and navigating items on the screen.
- **System Unit:** Contains key components like the motherboard, CPU, and memory.
- **Printer:** Produces hard copies of documents.

Meaning of GSM Repairs

GSM repairs refer to the **maintenance and fixing of mobile phones** and other communication gadgets that operate under the Global System for Mobile communication. GSM technology enables voice calls, SMS, and data transfer over cellular networks.

GSM repair involves diagnosing faults, replacing damaged parts, and ensuring that the phone functions properly. It requires knowledge of both hardware (physical parts) and software (programs that control operations).



Importance of Studying Computer Hardware and GSM Repairs

1. **Technological Literacy:** Helps students understand how computers and mobile phones work.
2. **Problem-Solving Skills:** Teaches how to identify and fix hardware or software faults.
3. **Employment Opportunities:** Provides valuable skills for working in repair centers, ICT firms, or self-employment.
4. **Self-Reliance:** Reduces dependence on others for repairs.
5. **Entrepreneurship Development:** Encourages students to start small businesses in phone and computer servicing.
6. **Support for Other Professions:** Enhances efficiency in offices, schools, and industries that rely on technology.

Career Opportunities in Computer Hardware and GSM Repairs

Studying this subject can open doors to several professions and business opportunities, including:

1. **Computer Technician:** Repairs and maintains computer systems.
2. **GSM Technician:** Repairs mobile phones and communication devices.
3. **Network Administrator:** Installs and manages computer networks.
4. **Hardware Engineer:** Designs and tests new computer parts.
5. **Sales Representative:** Buys and sells computers, phones, and accessories.
6. **ICT Consultant:** Advises individuals and organizations on technology use.
7. **Software Installer:** Installs operating systems and applications.
8. **Entrepreneur:** Owns and manages a computer or GSM repair business.

Relationship Between Computer Hardware and GSM Repairs

Computers and GSM devices share similar features, such as:

- Both contain hardware components like screens, batteries, and circuit boards.
- Both require software to function.
- Both use electricity and digital signals to operate.
- Both need regular maintenance and updates.

Therefore, skills learned in one area often help in the other, making them complementary fields.

Qualities of a Good Computer and GSM Repair Technician

A good technician must possess the following qualities:

1. **Patience:** Carefully examines faults before fixing them.
2. **Accuracy:** Avoids mistakes that may worsen the problem.
3. **Cleanliness:** Keeps the work area neat and organized.
4. **Honesty:** Builds trust with clients through transparent work.
5. **Curiosity:** Constantly learns about new technologies and devices.
6. **Safety Awareness:** Follows safety rules to prevent accidents.

Summary

Computer hardware and GSM repairs are vital fields in today's digital world. Computer hardware refers to the physical parts of a computer, while GSM repairs involve fixing mobile phones and similar devices. Learning these skills promotes technological literacy, self-reliance, and entrepreneurship. With continuous practice, students can build successful careers as technicians or ICT professionals.

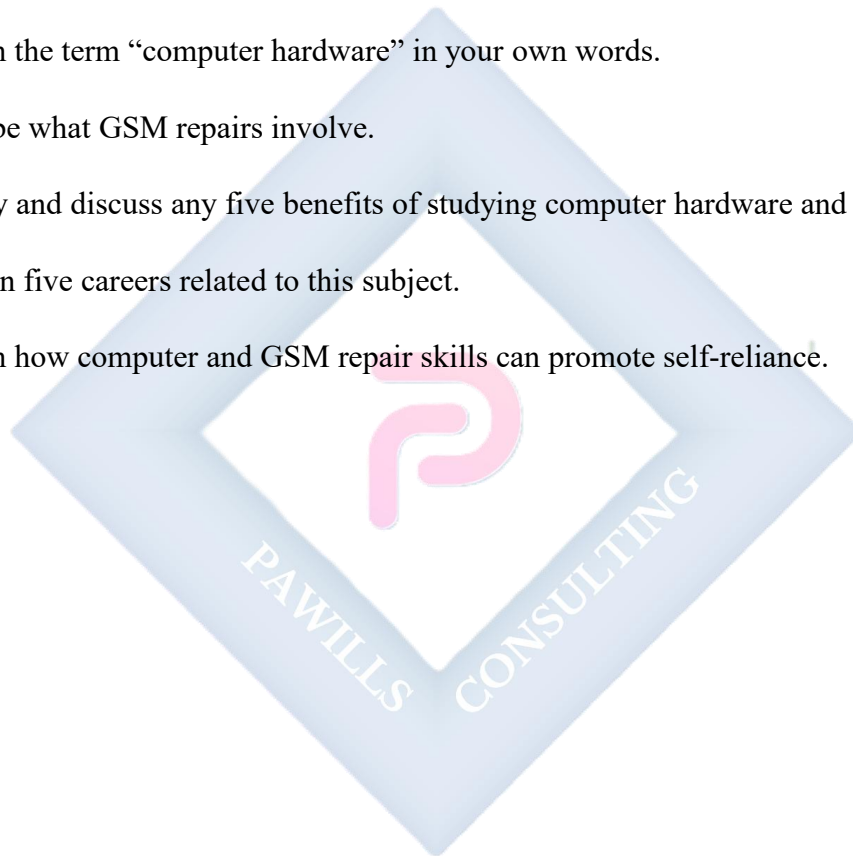
Evaluation

1. Define computer hardware.

2. What is GSM repair?
3. Mention three importance of learning computer hardware and GSM repairs.
4. List four career opportunities in this field.
5. State three qualities of a good technician.

Assignment

1. Explain the term “computer hardware” in your own words.
2. Describe what GSM repairs involve.
3. Identify and discuss any five benefits of studying computer hardware and GSM repairs.
4. Mention five careers related to this subject.
5. Explain how computer and GSM repair skills can promote self-reliance.



WEEK 2: HISTORY OF COMPUTERS AND GSM

Introduction

The history of computers and GSM (Global System for Mobile Communication) tells the story of how technology has evolved from simple devices into powerful tools that make our daily lives easier. Understanding this history helps us appreciate the growth of science and innovation that brought us the modern computer and mobile phone.

In this lesson, we will trace the development of computers and GSM, their inventors, and how they have changed the way people live, work, and communicate.



History of Computers

The development of computers did not happen overnight. It occurred over many years, through different generations, as technology improved.

1. Early Computing Devices (Before 1940)

The earliest form of computing began with **counting and calculation tools** such as:

- **Fingers and stones:** The first counting tools used by humans.
- **Abacus:** Invented about 5000 years ago; used by the Chinese and Egyptians for arithmetic operations.
- **Slide Rule:** Invented in the 1600s by William Oughtred, used for calculations.
- **Pascal's Calculator (1642):** Invented by Blaise Pascal to add and subtract numbers easily.
- **Charles Babbage's Analytical Engine (1833):** Considered the first concept of a modern computer.



2. Generations of Computers

Computers are grouped into **five generations** based on technological advancement.

Generation	Period	Main Technology	Features
First	1940–1956	Vacuum tubes	Large, slow, produced heat, used punch cards
Second	1956–1963	Transistors	Smaller, faster, more reliable
Third	1964–1971	Integrated Circuits (IC)	Increased speed and efficiency
Fourth	1971–Present	Microprocessors	Used in personal computers, portable and affordable
Fifth	Present and beyond	Artificial Intelligence (AI)	Uses voice recognition, robotics, and machine learning

3. Importance of Computers in Modern Times

1. Used in education for learning and research.
2. Essential in offices for document preparation and data storage.
3. Used in hospitals for medical record-keeping and diagnosis.
4. Important in communication and online services.
5. Used in entertainment, gaming, and social media.

Computers have evolved from large room-sized machines to lightweight laptops and tablets that fit into bags or pockets.

History of GSM (Global System for Mobile Communication)

1. Early Communication Systems

Before GSM technology, people used primitive communication methods:

- **Smoke signals and drums** were used for long-distance communication.

- **Postal letters** were common in ancient and medieval times.
- **Telegraph and telephone** were invented in the 19th century, marking the beginning of electronic communication.

2. Birth of GSM

GSM was introduced in the **1980s** as a digital standard for mobile communication. It replaced the older analog systems that were less secure and less efficient.

The first GSM network was launched in **Finland in 1991**, and since then, it has spread across the world, including Nigeria.

Key Points:

- GSM stands for **Global System for Mobile Communication**.
- It allows voice calls, SMS, and data transfer.
- It operates using **SIM cards**, which store user information.

3. Evolution of Mobile Phones

Generation	Period	Technology	Example
1G	1980s	Analog voice calls	Motorola DynaTAC
2G (GSM)	1990s	Digital voice & SMS	Nokia 3310
3G	2000s	Voice, video calls & internet	Blackberry, early smartphones
4G (LTE)	2010s	High-speed internet	iPhone, Android phones
5G	2020s	Super-fast connectivity & AI	Latest smartphones

Each generation improved speed, clarity, and connectivity.

4. Importance of GSM in Modern Society

1. Enables instant communication through calls and texts.

2. Allows access to the internet and social media.
3. Facilitates business through mobile banking and online marketing.
4. Provides entertainment through apps, videos, and music.
5. Supports education through e-learning and research.

GSM technology has revolutionized global communication, making the world more connected and efficient.

Similarities Between Computers and GSM Devices

1. Both use microprocessors to process data.
2. Both rely on software to function properly.
3. Both can connect to the internet.
4. Both require electricity or battery power.
5. Both can store and transfer information.

Summary

Computers and GSM have developed through centuries of technological progress. The computer evolved from simple counting tools like the abacus to advanced systems using artificial intelligence. Similarly, mobile phones have grown from basic voice devices to powerful smartphones capable of handling complex tasks. Both inventions have transformed communication, education, and business in our modern world.

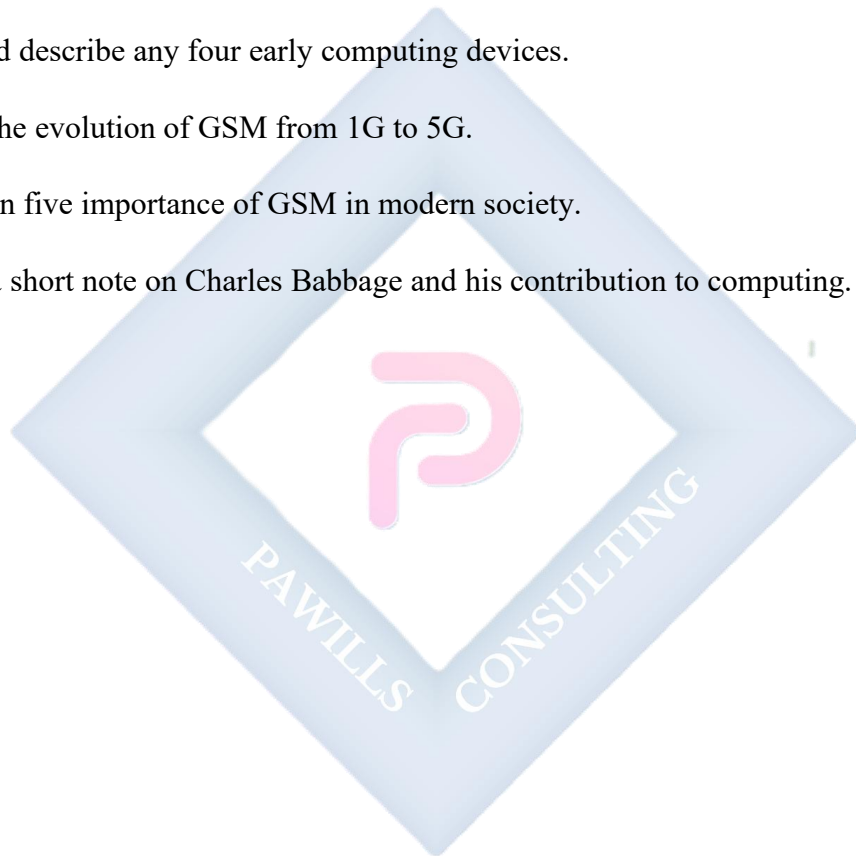
Evaluation

1. What does GSM stand for?
2. Mention the five generations of computers and their main features.

3. State three importance of computers in modern life.
4. Outline the main stages in the evolution of mobile communication.
5. Mention two similarities between computers and GSM devices.

Assignment

1. Explain the term “computer generation.”
2. List and describe any four early computing devices.
3. Trace the evolution of GSM from 1G to 5G.
4. Mention five importance of GSM in modern society.
5. Write a short note on Charles Babbage and his contribution to computing.



WEEK 3: BASIC COMPUTER HARDWARE COMPONENTS

Introduction

Every computer system is made up of two main parts: **hardware** and **software**. Hardware refers to the **physical, touchable parts** of the computer — the parts you can see and handle. Without hardware, a computer cannot function because it is the body that carries out the commands given by the software.

In this lesson, we will learn about the basic computer hardware components, their functions, and how they work together to make the computer operate effectively.



Meaning of Computer Hardware

Computer hardware refers to the **physical components or devices** that make up a computer system. These are the parts that can be seen and touched, such as the monitor, keyboard, mouse, CPU, printer, and so on.

In simple terms, hardware is the **machinery** of the computer that performs the physical tasks — processing, displaying, and storing data.

Examples of Computer Hardware:

Monitor, keyboard, mouse, system unit, printer, speaker, scanner, etc.

Main Categories of Computer Hardware Components

Computer hardware is grouped into **four main categories**:

1. **Input Devices**
2. **Output Devices**
3. **Processing Devices**
4. **Storage Devices**

Let's discuss each category in detail.

1. Input Devices

Input devices are used to **enter data and instructions** into the computer for processing. They allow users to communicate with the computer system.

Examples and Functions:

- **Keyboard:** Used to type letters, numbers, and commands.

- **Mouse:** Controls the movement of the pointer on the screen and is used to click, drag, or select items.
- **Scanner:** Converts printed documents and images into digital form.
- **Microphone:** Captures sound or voice input.
- **Webcam:** Captures live images or videos for online meetings or calls.

2. Output Devices

Output devices display the **results of processed data** to the user. They convert digital information into a form humans can understand — text, sound, or images.

Examples and Functions:

- **Monitor:** Displays text, graphics, and videos.
- **Printer:** Produces a hard copy (paper output) of documents and images.
- **Speaker:** Produces sound output such as music, alarms, or spoken words.
- **Projector:** Displays enlarged visuals on a screen for presentations.
- **Plotter:** Prints large drawings such as maps or engineering designs.

3. Processing Devices

Processing devices perform the **actual work** of computing by carrying out arithmetic and logical operations on data. The central processing device is the **CPU**.

Central Processing Unit (CPU):

The CPU (Central Processing Unit) is known as the “**brain**” of the computer. It controls all other components and processes data entered into the system.

The CPU is divided into three parts:

- **Arithmetic and Logic Unit (ALU):** Performs all mathematical and logical operations.

- **Control Unit (CU):** Directs the operation of all other units by following instructions.
- **Memory Unit:** Temporarily stores data and instructions while processing.

4. Storage Devices

Storage devices are used to **store data and programs** permanently or temporarily for future use. They are divided into **primary storage** and **secondary storage**.

Primary Storage (Main Memory):

- Located inside the computer.
- Used for temporary data storage while the computer is running.
- Examples: **RAM (Random Access Memory)** and **ROM (Read Only Memory)**.

Secondary Storage:

- Used for permanent data storage.
- Retains information even when the computer is off.
- Examples: **Hard disk, flash drive, CD/DVD, external drives, memory cards**.

Relationship Between the Components

All computer hardware components work together as a system:

1. **Input devices** collect data and send it to the computer.
2. **Processing devices (CPU)** handle the data and perform operations.
3. **Output devices** present the results to the user.
4. **Storage devices** save the data for future use.

For example, when you type a letter:

- The keyboard (input) sends your text to the CPU (processing).
- The CPU processes it and displays it on the monitor (output).

- The document can be saved on the hard disk (storage).

Importance of Computer Hardware

1. Enables the computer to perform physical functions.
2. Makes data input, processing, and output possible.
3. Supports communication between user and computer.
4. Provides storage for information and software.
5. Determines the computer's speed and efficiency.

Care and Maintenance of Computer Hardware

1. Keep the computer clean and dust-free.
2. Avoid eating or drinking near the computer.
3. Switch off the computer properly after use.
4. Handle all cables and ports gently.
5. Protect the computer from power surges using a stabilizer or UPS.

Summary

Computer hardware refers to all the physical parts of the computer system. These include input, output, processing, and storage devices. Each category has its own role, but together, they help the computer perform complex tasks efficiently. Taking proper care of hardware ensures that the computer works effectively and lasts longer.

Evaluation

1. Define computer hardware.
2. List and explain the four categories of computer hardware.
3. What are two examples each of input and output devices?
4. Mention the three main parts of the CPU and their functions.
5. State three ways of maintaining computer hardware.

Assignment

1. Draw and label a typical computer system.
2. Explain the difference between primary and secondary storage.
3. State five examples of input devices and their uses.
4. Write short notes on the functions of the CPU.
5. Mention four precautions to take when handling computer hardware.

WEEK 4: IDENTIFICATION OF GSM COMPONENTS

Introduction

GSM (Global System for Mobile Communication) is one of the most widely used technologies for mobile phones all over the world. Every GSM phone, whether simple or sophisticated, is made up of different components that work together to perform various functions such as making calls, sending messages, browsing the internet, and storing information.

In this lesson, we shall learn about the major components of a GSM phone, their functions, and how to identify them physically.



Meaning of GSM Components

GSM components are the **individual parts and accessories** that make up a mobile phone. These parts are either **internal** (found inside the phone) or **external** (found outside the phone).

Each component has a specific function that helps the phone to operate properly. When one component is faulty, it can affect the entire phone's performance.

Classification of GSM Components

GSM phone components are divided into two major categories:

1. **External (Visible) Components**
2. **Internal (Invisible) Components**

Let's examine each group in detail.

1. External Components of a GSM Phone

These are the parts of the phone that can be **seen and touched** easily. They are used directly by the user to operate the phone.

Main External Components and Their Functions:

- **Screen (Display Unit):**
The screen displays information such as text messages, pictures, time, videos, and applications. Modern phones have touchscreens that respond to finger gestures.
- **Keypad/Touchpad:**
This is used to enter numbers, letters, and commands. Older phones have physical keypads, while smartphones use touch keypads.

- **Battery:**

The battery supplies electrical power to the phone. It is rechargeable and determines how long the phone can be used before recharging.

- **Back Cover/Phone Casing:**

This protects the internal parts of the phone from dust, water, and physical damage.

- **Camera Lens:**

Used for capturing photos and videos. Some phones have both front and back cameras.

- **Earpiece/Speaker:**

Produces sound when making calls or playing music.

- **Charging Port:**

The point where the charger is inserted to power or charge the battery.

- **SIM Card Slot:**

The compartment where the SIM card is inserted. It connects the phone to the mobile network.

- **Memory Card Slot:**

A space where an external memory card is inserted to store more files and data.

2. Internal Components of a GSM Phone

These components are located **inside the phone** and are usually covered by the phone casing. They are responsible for the actual operation and processing of signals.

Main Internal Components and Their Functions:

- **Circuit Board (Motherboard):**

This is the heart of the phone. It holds all the major electronic components that control the phone's functions.

- **SIM Card:**

A small chip inserted into the phone to connect it to the mobile network. It stores information such as phone number, network, and contacts.

- **Antenna:**
Helps to receive and transmit radio signals between the phone and the network tower for communication.
- **Microphone:**
Captures the user's voice when making calls or recording audio.
- **Vibrator Motor:**
Responsible for vibrating the phone during silent notifications or incoming calls.
- **Ringer/Buzzer:**
Produces alert sounds for incoming calls or messages.
- **Display Connector (Flex):**
Connects the screen to the circuit board, transmitting display signals.
- **Power IC (Integrated Circuit):**
Controls the power distribution within the phone.
- **Memory Chip:**
Stores applications, system files, and user data.
- **Charging IC:**
Manages charging current from the charger to the battery.

Functions of Major GSM Components

Component	Function
Screen	Displays text, pictures, and videos.
Keypad/Touchpad	Allows user input for typing and dialing.
Battery	Supplies power to the phone.
SIM Card	Connects to mobile networks and stores contact info.

Circuit Board	Controls phone operation and connects all components.
Antenna	Receives and transmits network signals.
Microphone	Captures sound during calls.
Speaker	Produces audio output.
Camera	Takes photos and videos.
Charger Port	Enables battery charging.

Importance of Understanding GSM Components

1. Helps in identifying and repairing faulty parts.
2. Aids in proper phone maintenance.
3. Encourages responsible usage and handling.
4. Promotes technical skills for future careers in GSM repairs.
5. Prevents damage caused by mishandling or ignorance.

Care and Handling of GSM Components

1. Always switch off the phone before removing the battery or SIM card.
2. Avoid exposing the phone to water or extreme heat.
3. Do not drop the phone or press it too hard.
4. Keep the charging port and headphone jack clean.
5. Use only recommended accessories for your phone model.

Summary

GSM components are the various parts that make up a mobile phone. They are divided into **external components** (such as the screen, keypad, and battery) and **internal components** (such as the circuit board, SIM card, and antenna). Each component has a unique role in ensuring the phone functions properly. Proper knowledge of these parts helps in using, maintaining, and repairing mobile phones effectively.

Evaluation

1. What is meant by GSM components?
2. List five external components of a GSM phone.
3. Mention any four internal components and their functions.
4. What is the function of the antenna in a GSM phone?
5. State three ways of caring for a GSM phone.

Assignment

1. Explain the difference between internal and external GSM components.
2. Draw and label a simple diagram of a GSM phone showing its main parts.
3. List and explain five internal components and their functions.
4. State five precautions to observe when handling GSM components.
5. Identify three components that help in sound production and explain their roles.

WEEK 6: SAFETY PRECAUTIONS IN WORKSHOP

Introduction

The workshop is a place where practical repair and maintenance activities on computers and GSM phones are carried out. Because this environment involves the use of **electrical equipment, sharp tools, soldering irons, and delicate devices**, safety precautions must be observed at all times to prevent injuries and damage.

This lesson focuses on understanding **workshop safety precautions**, how to handle **tools and equipment safely**, and the importance of maintaining a **clean, organized, and hazard-free work area**.

Meaning of Workshop Safety

Workshop safety refers to the set of **rules, actions, and habits** that ensure protection from accidents or injuries while working in a workshop. It also involves keeping both people and equipment safe from harm or damage.

In other words, it is the **discipline of working carefully and responsibly** in an environment where electrical, chemical, or mechanical tools are used.



Importance of Safety Precautions in the Workshop

1. **Prevents Accidents and Injuries:** Following safety rules protects workers from burns, cuts, electric shocks, and other injuries.
2. **Protects Tools and Equipment:** Careful handling prevents damage to delicate devices like circuit boards or screens.
3. **Ensures Efficiency:** A safe, organized workspace allows technicians to work faster and more effectively.
4. **Reduces Cost of Repairs:** Prevents unnecessary damage and reduces money spent on replacements.
5. **Promotes Professionalism:** Observing safety rules builds discipline and confidence among workers.

6. **Encourages Team Safety:** Everyone in the workshop remains safe when all members follow precautions.
7. **Maintains Good Health:** Safety reduces exposure to harmful fumes and electrical hazards.



8.

Common Hazards in the Workshop

Workshop hazards are things that can cause harm if not properly managed. They include:

1. **Electrical Hazards:** From faulty wiring, open sockets, or wet hands when using electrical tools.
2. **Sharp Tools:** Blades, screwdrivers, and pliers can cause cuts if carelessly handled.
3. **Hot Equipment:** Soldering irons, heating elements, and chargers can cause burns.
4. **Slippery Floors:** Spilled water, oil, or dust can cause falls.

5. **Improper Ventilation:** Leads to inhaling fumes from soldering or cleaning chemicals.
6. **Overcrowded Workspaces:** Can cause collisions and breakages.
7. **Cluttered Workbenches:** Increases risk of accidents and loss of tools.

General Workshop Safety Precautions

To maintain safety in the workshop, all technicians and students must observe the following general precautions:

1. **Wear Appropriate Clothing:**
Avoid loose clothing, dangling jewelry, or long hair that could get caught in equipment.
Wear overalls or lab coats.
2. **Use Personal Protective Equipment (PPE):**
Always use gloves, nose masks, safety goggles, and closed shoes to protect yourself.
3. **Keep the Workshop Clean and Organized:**
Sweep regularly, arrange tools properly, and clean spills immediately.
4. **Avoid Food and Drinks in the Workshop:**
Eating or drinking near electrical equipment can cause contamination or electric shock.
5. **Do Not Operate Faulty Equipment:**
Report damaged or malfunctioning tools to the instructor for repair or replacement.
6. **Disconnect Power Before Repairs:**
Always switch off and unplug devices before opening or repairing them.
7. **Handle Soldering Irons Carefully:**
Keep hot soldering irons in their stands when not in use to prevent burns.
8. **Avoid Overloading Electrical Sockets:**
Plug only necessary devices into power outlets to prevent short circuits or fire.
9. **Use Tools Properly:**
Handle all tools as instructed. Do not misuse them or apply unnecessary force.

10. Maintain Good Ventilation:

Open windows or use exhaust fans when soldering or using chemicals to avoid inhaling fumes.

Handling Tools Safely

1. Always check that tools are in good condition before use.
2. Use the right tool for the right job.
3. Return tools to their proper place after use.
4. Hold tools by their insulated or handle parts, especially electrical ones.
5. Avoid lending tools carelessly or throwing them around.

Electrical Safety Rules

Since computer and GSM repairs involve electricity, observe these specific precautions:

1. Never touch electrical appliances with wet hands.
2. Use only properly insulated wires and plugs.
3. Avoid working on live circuits; always disconnect power.
4. Do not overload extension sockets.
5. Always keep liquids away from electrical devices.
6. Wear rubber-soled shoes when working with electricity.
7. Replace damaged cords or plugs immediately.
8. Use a voltage stabilizer or UPS to protect equipment.

ESD Safety (Electrostatic Discharge)

Electrostatic discharge (ESD) is the sudden flow of electricity between two objects that can damage sensitive electronic components like circuit boards and microchips.

To prevent ESD:

1. Use anti-static wrist straps when handling internal phone or computer parts.
2. Work on anti-static mats.
3. Avoid wearing wool or nylon clothes that create static electricity.
4. Ground yourself before touching internal components.
5. Keep electronic parts in anti-static bags.

Emergency Measures in the Workshop

In case of an accident:

1. Switch off all electrical appliances immediately.
2. Move the injured person away from danger.
3. Do not touch anyone receiving an electric shock directly—use a wooden stick or non-metallic object.
4. Call for help and report to a teacher or supervisor.
5. Administer first aid before seeking medical assistance.

Summary

Workshop safety is essential to prevent accidents, injuries, and equipment damage. It involves wearing protective gear, handling tools correctly, observing electrical precautions, and maintaining cleanliness and order. A safe and organized workshop ensures efficiency, professionalism, and long-term productivity for all technicians.

Evaluation

1. Define workshop safety.
2. List five common workshop hazards.
3. Mention five general safety precautions in the workshop.
4. Explain what ESD means and give two ways of preventing it.
5. State three emergency measures to take when an accident occurs in the workshop.

Assignment

1. Explain why safety is important in a computer and GSM repair workshop.
2. List and describe five personal protective equipment used in the workshop.
3. Identify four electrical safety precautions.
4. Explain three ways to handle tools safely.
5. Describe how to prevent ESD damage when handling GSM circuit boards.

WEEK 8: TOOLS FOR COMPUTER AND GSM REPAIRS

Introduction

Tools are essential instruments used by technicians to carry out specific tasks in computer and GSM repair work. Without the right tools, repairs cannot be done properly, efficiently, or safely. A good technician must be familiar with the **names, functions, and proper handling** of all repair tools.

In this lesson, we shall study the various tools used in computer and GSM repairs, their types, uses, and safety precautions in handling them.

Meaning of Tools

A **tool** is any instrument or device used to carry out a particular task, especially in mechanical, electrical, or electronic work. Tools help the technician to open, fix, test, or assemble devices with ease and accuracy.

In computer and GSM repairs, tools are used to:

- Open and assemble components.
- Solder or desolder electronic parts.
- Test for faults and electrical flow.
- Handle small and delicate components safely.



Types of Tools Used in Computer and GSM Repairs

The tools can be grouped into the following categories:

1. **Mechanical Tools**
2. **Electrical Tools**
3. **Testing and Measuring Tools**
4. **Specialized GSM Tools**

Let us discuss each group in detail.

1. Mechanical Tools

These are hand tools used for loosening, tightening, or holding components during repairs.

Examples and Uses:

- **Screwdrivers:** Used for loosening and tightening screws of different sizes.
 - *Types:* Flat-head, star (Phillips), and precision screwdrivers.

- **Pliers:** Used to hold, bend, or cut wires and small components.
 - *Types:* Long-nose pliers, combination pliers, and cutting pliers.
- **Tweezers:** Used to pick or hold very small electronic parts like resistors or chips.
- **Spanners and Nut Drivers:** Used to tighten or loosen nuts and bolts in computer casings.
- **Brush:** Used for cleaning dust from keyboards, fans, and circuit boards.
- **Blower or Air Duster:** Removes dust and particles from computer and phone parts.

2. Electrical Tools

These are tools that use electricity to function. They are mainly used for soldering, testing, and repairs of electrical circuits.

Examples and Uses:

- **Soldering Iron:** Used to melt solder for joining wires and components on circuit boards.
- **Soldering Station:** A complete unit with adjustable temperature for safe soldering.
- **Desoldering Pump (Solder Sucker):** Used to remove melted solder during component replacement.
- **Hot Air Gun (Rework Station):** Used to remove or replace surface-mounted components on circuit boards.
- **Electric Drill:** Used to create small holes in casings or boards when necessary.
- **Magnifying Lamp:** Provides bright light and magnification for viewing tiny parts.



3. Testing and Measuring Tools

These tools are used to test voltage, resistance, continuity, and other electrical values in devices. They help in detecting faults in GSM and computer systems.

Examples and Uses:

- **Multimeter:** Measures voltage, resistance, and current in a circuit.
- **Continuity Tester:** Checks if electrical current can flow through a wire or component.
- **Power Supply Unit:** Provides adjustable power for testing devices under repair.
- **Battery Tester:** Used to check the voltage and strength of phone batteries.

- **Signal Tracer:** Helps in tracing signals and identifying faulty lines on a circuit board.

4. Specialized GSM Tools

These are tools specifically designed for mobile phone repair.

Examples and Uses:

- **SIM Card Reader/Writer:** Used to read or copy data from SIM cards.
- **Phone Opener Kit:** Includes small prying tools and suction cups for opening phone casings safely.
- **PCB Holder/Clamp:** Holds the phone's circuit board in place during soldering.
- **Anti-Static Wrist Strap:** Prevents electrostatic discharge that can damage phone components.
- **Spudger:** A flat, plastic tool used to separate screen and casing without damage.
- **Heat Mat:** Protects the work surface from heat during soldering or hot-air repairs.

Safety Precautions When Using Tools

1. Always wear safety gear such as gloves, goggles, and masks.
2. Switch off electrical tools when not in use.
3. Handle hot tools like soldering irons carefully to avoid burns.
4. Keep tools clean and dry to prevent rust and electric shock.
5. Use the correct tool for each task.
6. Disconnect electrical equipment from power supply before repairing.
7. Do not overload sockets with multiple tools.
8. Store all tools properly after use.

9. Keep liquids away from electrical tools.
10. Inspect tools regularly for damage and replace faulty ones.

Maintenance of Tools

Proper maintenance ensures the tools last long and remain effective.

1. Clean tools after every use.
2. Oil metallic parts to prevent rusting.
3. Store tools in dry, safe toolboxes or cabinets.
4. Replace worn-out or blunt tools immediately.
5. Keep electrical tools away from moisture and dust.
6. Calibrate measuring tools like multimeters periodically.
7. Handle precision tools with care to avoid misalignment.

Summary

Tools are indispensable in computer and GSM repair work. They are classified into **mechanical, electrical, testing, and specialized GSM tools**. Each tool has a specific function that helps the technician perform accurate and efficient repairs. Observing safety rules and maintaining tools properly ensures durability and professional results.

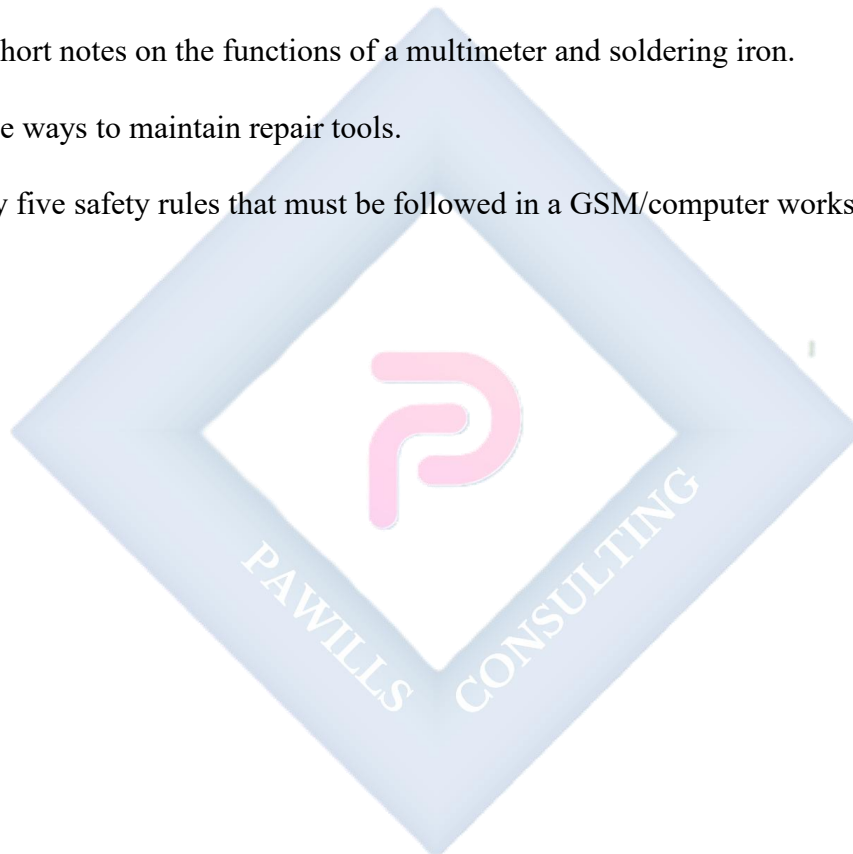
Evaluation

1. What is a tool?
2. List and explain four types of tools used in computer and GSM repairs.
3. Mention five examples of mechanical tools and their uses.

4. What are testing tools used for?
5. State three safety precautions when using electrical tools.

Assignment

1. Define tools and explain their importance in GSM repair.
2. Classify tools into four main groups with two examples each.
3. Write short notes on the functions of a multimeter and soldering iron.
4. List five ways to maintain repair tools.
5. Identify five safety rules that must be followed in a GSM/computer workshop.

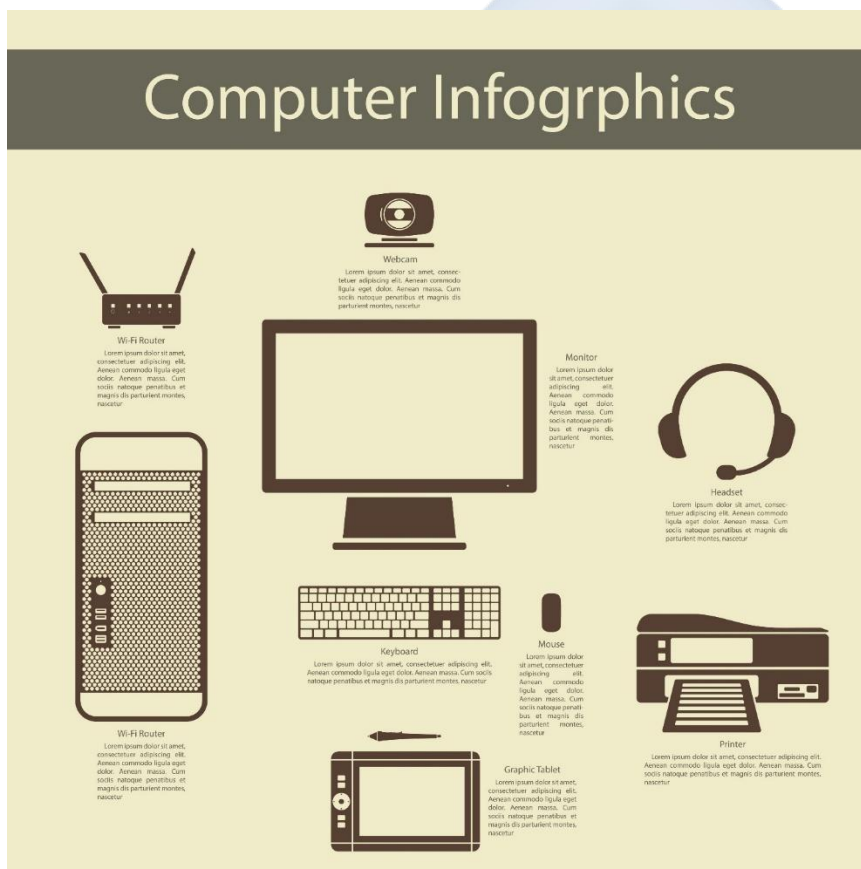


WEEK 9: PERIPHERAL DEVICES

Introduction

Peripheral devices are external components connected to a computer or GSM system to enhance its functionality. They allow users to input, output, or store data and to communicate with other systems. Without peripherals, computers and GSM devices would have very limited usefulness because they extend the capability of the main system unit.

In this lesson, we will learn the **meaning, types, functions, and examples** of peripheral devices used in computer systems and mobile technology.

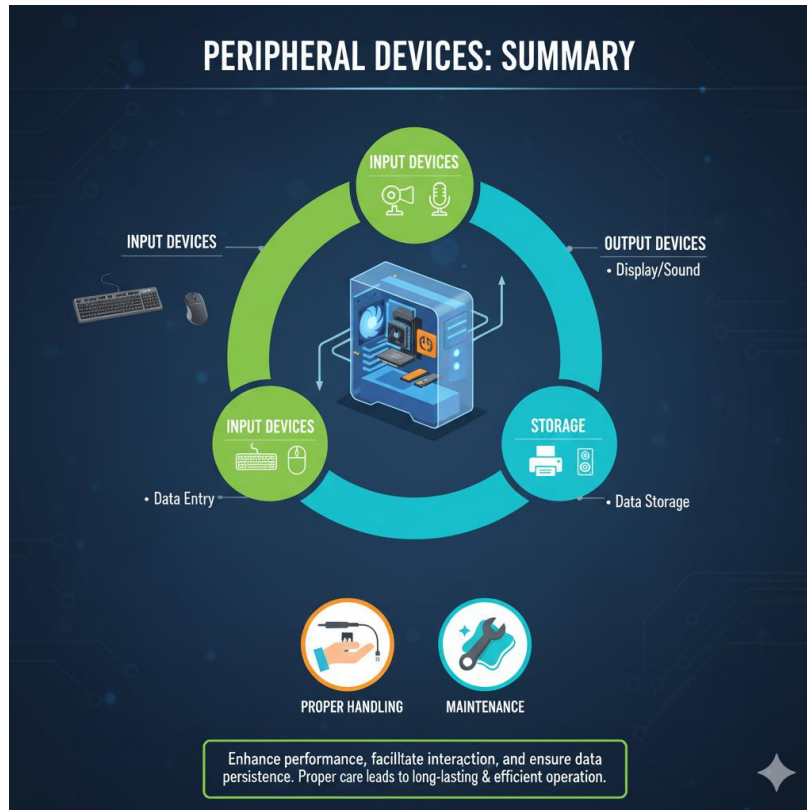


Meaning of Peripheral Devices

A **peripheral device** is any hardware component that is **connected externally** or **attached to the main computer system unit** to perform specific tasks. These devices are not part of the

computer's core architecture (CPU, RAM, motherboard, etc.), but they assist the computer in receiving, storing, or delivering information.

For example, a **printer** prints documents from the computer, while a **scanner** sends pictures or text from paper into the computer.



Types of Peripheral Devices

Peripheral devices are grouped into **three main categories** based on their functions:

1. **Input Devices**
2. **Output Devices**
3. **Storage Devices**

Each category has distinct functions that support computer or GSM operations.

1. Input Devices

Input devices are peripherals used to send data or commands into the computer. They allow the user to interact with the system.

Common Examples:

- **Keyboard:** Used to type letters, numbers, and commands into the computer.
- **Mouse:** A pointing device used to move the cursor and select items on the screen.
- **Scanner:** Converts hard copy documents and images into digital form.
- **Microphone:** Captures sound or voice input into the computer.
- **Digital Camera or Webcam:** Captures still or moving images.
- **Touch Screen:** Combines display and input functions, allowing users to touch the screen to control the device.
- **Barcode Reader:** Scans product codes in shops and libraries.

2. Output Devices

Output devices are peripherals that display, project, or produce the results of computer processing. They allow the computer to communicate information to the user.

Common Examples:

- **Monitor:** Displays images, videos, and text from the computer.
- **Printer:** Produces hard copies of documents and pictures.
- **Speaker:** Produces sound output from the computer or phone.
- **Headphones or Earphones:** Provide private listening of audio output.
- **Projector:** Displays enlarged visuals for classroom or business presentations.
- **Plotter:** Used to produce large drawings such as engineering plans and architectural designs.

3. Storage Devices

Storage devices are used to **save, retrieve, and transfer data** from one device to another. They can be internal (inside the system unit) or external (connected through USB or other ports).

Common Examples:

- **Flash Drive (USB Drive):** Portable device for transferring and storing data.
- **External Hard Drive:** Used for backing up large amounts of files and programs.
- **Memory Card (SD Card):** Used in mobile phones, cameras, and tablets for data storage.
- **Compact Disc (CD) and Digital Versatile Disc (DVD):** Used to store music, videos, and software.
- **External Solid-State Drive (SSD):** High-speed data storage used in modern computers.
- **Cloud Storage:** Internet-based storage service such as Google Drive or iCloud.

Peripheral Devices in GSM Systems

Mobile phones also use peripherals similar to computers. These peripherals enhance phone performance and expand functionality.

Examples:

- **Charger:** Provides electrical power to the phone battery.
- **Earphones or Bluetooth Headset:** For audio input and output.
- **USB Cable:** Used for transferring data between phone and computer.
- **Memory Card:** Stores photos, videos, and documents.
- **Power Bank:** Portable power device for charging phones.
- **Bluetooth Speaker:** Plays music wirelessly from the phone.
- **Smartwatch:** Connects to the phone to display notifications and track fitness activities.

Importance of Peripheral Devices

Peripheral devices are essential for several reasons:

1. **Enhance Functionality:** They add new capabilities to computers and phones.
2. **Ease of Operation:** They make it easier for users to communicate with the system.
3. **Data Storage and Transfer:** They allow safe and quick data movement between systems.
4. **Productivity:** Enable users to print, scan, record, or present efficiently.
5. **Entertainment:** Provide audio, video, and gaming experiences.
6. **Connectivity:** Facilitate sharing of data and internet access.

Care and Maintenance of Peripheral Devices

1. Handle all peripherals carefully and avoid dropping them.
2. Keep devices away from water and dust.
3. Clean monitors, keyboards, and printers regularly using soft, dry cloths.
4. Eject flash drives safely before removal.
5. Avoid overloading USB ports with multiple devices.
6. Store removable devices like memory cards and flash drives in cases.
7. Keep cables neatly arranged to prevent damage.
8. Disconnect power supply before cleaning.
9. Update drivers regularly for modern devices.
10. Use surge protectors to guard against electrical damage.

Summary

Peripheral devices are external components connected to computers or GSM systems to enhance their performance. They are classified into **input**, **output**, and **storage** devices. Each performs unique roles such as data entry, display, and storage. Proper handling and maintenance ensure long-lasting and efficient operation.

Evaluation

1. What is a peripheral device?
2. List the three main types of peripheral devices.
3. Give three examples each of input, output, and storage devices.
4. Mention five peripherals used in GSM systems.
5. State three ways to care for peripheral devices.

Assignment

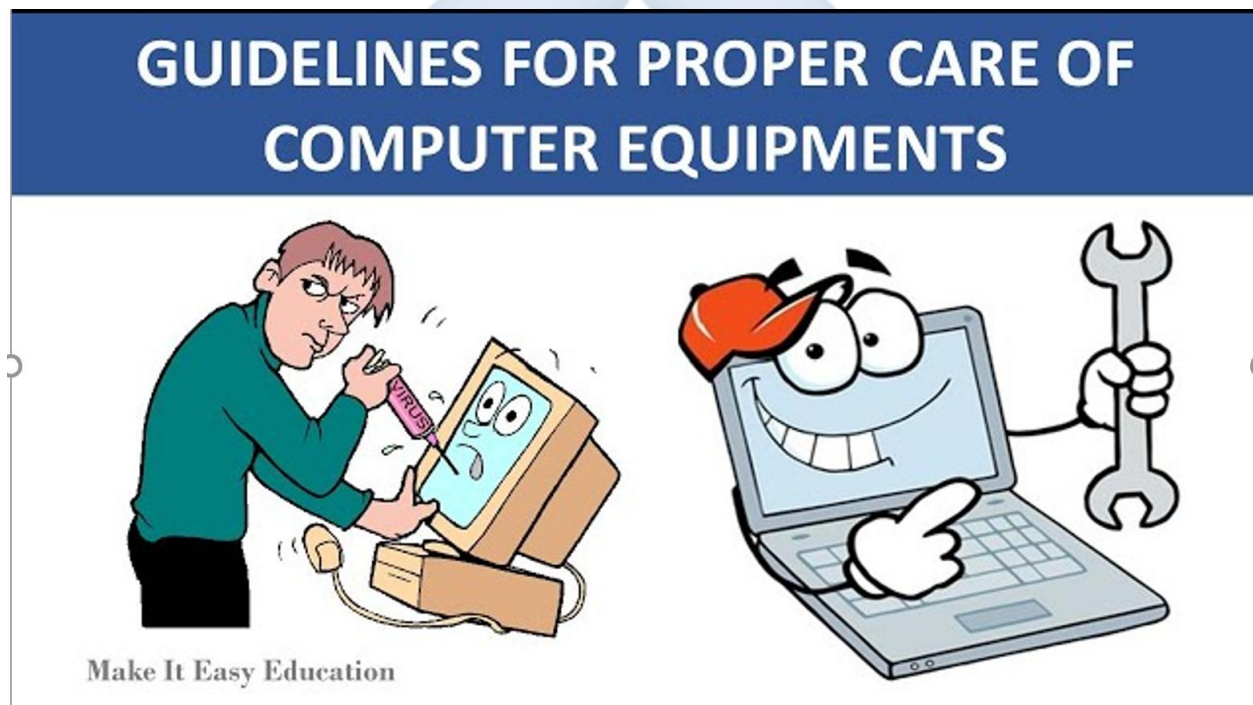
1. Define peripheral devices and state their importance.
2. Draw and label any three input devices.
3. List and explain four examples of storage devices.
4. Identify five common peripheral devices used with mobile phones.
5. Explain three maintenance practices for keeping peripherals in good condition.

WEEK 10: BASIC CARE AND MAINTENANCE OF COMPUTER AND GSM

Introduction

Every machine or device, no matter how strong, needs regular care and maintenance to keep it functioning properly. Computers and GSM (Global System for Mobile communication) devices are no exception. Without proper maintenance, they can easily develop faults, lose data, or stop working altogether.

This lesson explains the **meaning of maintenance**, **types of maintenance**, **importance of care**, and **practical ways to care for computers and GSM devices** to make them last longer and perform efficiently.



Meaning of Maintenance

Maintenance means the act of keeping something in good condition through regular cleaning, inspection, repair, and servicing. In computers and GSM systems, maintenance involves taking steps to prevent faults, detect problems early, and repair or replace damaged components.

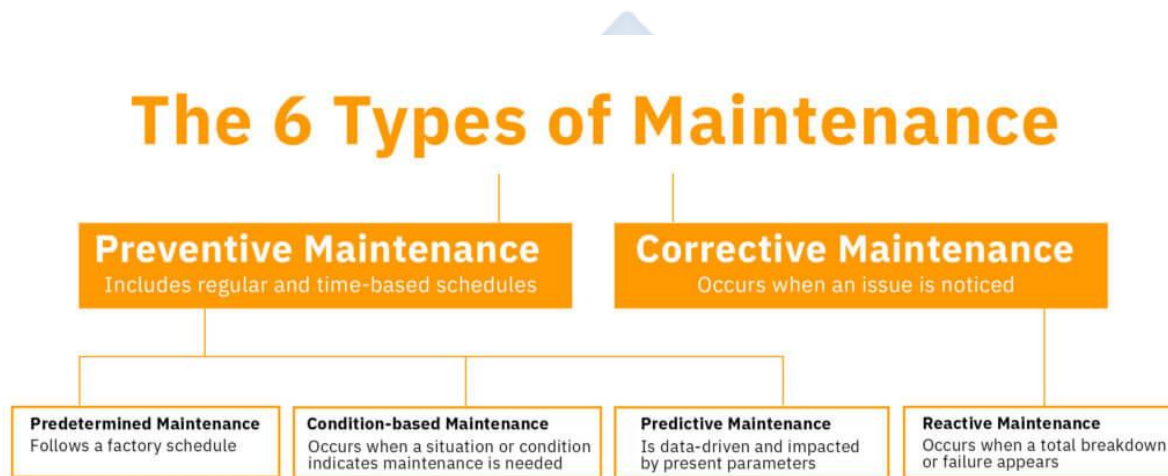
It helps ensure that the devices work smoothly and efficiently at all times.

Types of Maintenance

There are **two major types** of maintenance for computers and GSM devices:

1. **Preventive Maintenance**
2. **Corrective Maintenance**

Let's look at each type in detail.



1. Preventive Maintenance

Preventive maintenance refers to **regular servicing and cleaning** of devices to **avoid future problems or breakdowns**.

It involves simple actions like cleaning, dusting, checking cables, charging batteries properly, and updating software.

Examples:

- Regularly cleaning the computer keyboard and screen.
- Ensuring proper ventilation around the computer.
- Scanning the system for viruses.
- Avoiding overcharging the phone battery.

- Storing phones and computers away from dust and moisture.

2. Corrective Maintenance

Corrective maintenance involves **repairing or replacing faulty parts** after a problem has occurred. It is done when a computer or phone stops working properly or shows signs of malfunction.

Examples:

- Replacing a broken computer screen.
- Repairing a faulty phone charging port.
- Changing a dead laptop battery.
- Fixing damaged software or reinstalling the operating system.

Importance of Maintenance

Maintenance is very important because it:

1. **Prevents Damage:** Regular care prevents major faults and breakdowns.
2. **Saves Money:** Helps avoid costly repairs or replacements.
3. **Improves Performance:** Ensures computers and phones run faster and more efficiently.
4. **Extends Lifespan:** Well-maintained devices last longer.
5. **Protects Data:** Prevents loss of files due to sudden hardware failure.
6. **Ensures Safety:** Reduces risks of electric shock or overheating.
7. **Enhances Productivity:** Keeps devices always ready for use.

Basic Care for Computer Systems

To keep a computer in good condition, the following care practices should be observed:



1. Regular Cleaning:

Clean the monitor, keyboard, and system unit with a soft, dry cloth. Avoid using water directly.

2. Proper Ventilation:

Keep the system in a well-ventilated area to prevent overheating.

3. Use of Power Surge Protectors:

Protect the computer from voltage fluctuations or power surges.

4. Avoid Eating or Drinking Near the Computer:

Food particles and liquids can damage internal parts.

5. **Install Antivirus Software:**

Protects the computer from malware and viruses that can corrupt data.

6. **Shut Down Properly:**

Always use the shutdown option; don't unplug abruptly.

7. **Regular Backups:**

Save important files on external drives or cloud storage.

8. **Handle with Care:**

Avoid shaking, dropping, or moving the system while it's running.

9. **Use Uninterruptible Power Supply (UPS):**

Helps keep the computer running when there's a sudden power outage.

10. **Avoid Overloading the System:**

Do not run too many applications at once.

Basic Care for GSM Devices

Mobile phones and tablets also need proper maintenance to function effectively. The following practices ensure good performance and longer lifespan:

1. **Charge Properly:**

Avoid overcharging or letting the battery drain completely before charging.

2. **Clean Regularly:**

Wipe the screen and body with a soft, dry cloth to remove dust and fingerprints.

3. **Use Screen Guards and Phone Cases:**

Protects the phone from scratches, cracks, and damage when dropped.

4. **Avoid Moisture:**

Keep the phone away from water or high humidity areas.

5. Handle with Care:

Do not sit or sleep on the phone. Handle it gently to avoid damage to the screen and internal parts.

6. Update Software Regularly:

Install system updates to improve performance and security.

7. Avoid Installing Harmful Apps:

Only download apps from trusted sources like Google Play Store or Apple App Store.

8. Don't Expose to Heat:

Keep the phone away from direct sunlight and heat sources.

9. Use Original Accessories:

Always use the manufacturer's recommended chargers and earphones.

10. Store Properly:

When not in use, keep the phone in a safe and dry place.

Common Maintenance Tools

Technicians use some tools to carry out maintenance activities effectively. These include:

- Soft brush and blower for dust removal.
- Screwdrivers for opening system units or phones.
- Multimeter for testing electrical components.
- Cleaning solution for computer parts.
- Soldering iron for fixing loose components.
- Anti-static wrist strap to prevent electrical shock.

Signs That Maintenance Is Needed

1. The computer or phone heats up quickly.

2. The system becomes unusually slow.
3. The battery drains too fast.
4. The screen flickers or displays errors.
5. Strange sounds come from the system unit or phone.
6. The device refuses to power on or crashes frequently.

When these signs appear, maintenance should be carried out immediately to prevent more serious damage.

Summary

Maintenance means taking care of computers and GSM devices to keep them in good working condition. It is classified into **preventive** and **corrective maintenance**. Regular maintenance prevents faults, saves money, and extends the lifespan of devices. Both computers and GSMs require proper cleaning, careful handling, regular software updates, and the use of correct accessories for optimal performance.

Evaluation

1. What is maintenance?
2. State the two main types of maintenance.
3. Give four examples of preventive maintenance activities.
4. Mention three ways to care for a GSM device.
5. List three importance of maintenance.

Assignment

1. Define preventive and corrective maintenance, giving two examples of each.

2. Mention five common tools used in maintaining computers and GSM devices.
3. Explain three ways to protect your computer from damage.
4. Describe three signs that indicate your phone needs maintenance.
5. List five benefits of maintaining your computer and GSM devices regularly.



WEEK 11: REVISION (REVIEW OF MAJOR TOPICS)

Introduction

Revision is the process of going over what has been taught in the term to ensure proper understanding, retention, and readiness for assessment. It allows students to recall key concepts, definitions, and practices learned in previous weeks. In this final week of the first term, we will review all the major topics covered under **Computer Hardware and GSM Repairs**.

This revision will help students consolidate their knowledge and prepare for the end-of-term examination.

Summary of Key Topics

1. Introduction to Computer Hardware and GSM Repairs

- **Computer Hardware:** The physical parts of a computer that can be touched and seen, such as the monitor, keyboard, mouse, and CPU.
- **GSM Repairs:** The act of fixing and maintaining mobile phones that use the Global System for Mobile communication.
- **Importance:** Promotes self-reliance, job creation, and technical skills development.

2. History of Computers and GSM

- **History of Computers:** The evolution of computers from early counting devices (like the abacus) to modern digital computers.
 - Generations: Mechanical, Electromechanical, Electronic (1st to 5th generation).
- **History of GSM:** Started from wired telephones to mobile cellular systems.
 - Nigeria adopted GSM technology in 2001.
- **Pioneers:** Charles Babbage (Father of the Computer), Alexander Graham Bell (Telephone), and Martin Cooper (Mobile Phone).

3. Basic Computer Hardware Components

- **Input Devices:** Keyboard, mouse, scanner.
- **Output Devices:** Monitor, printer, speaker.
- **Processing Devices:** Central Processing Unit (CPU).
- **Storage Devices:** Hard drive, flash drive, memory card.
- **System Unit:** Houses the motherboard and processor.

4. Identification of GSM Components

- **Common parts:** Screen, keypad, battery, SIM slot, charging port, circuit board, and antenna.
- **Functions:** The screen displays information, while the battery supplies power.

5. Safety Precautions in the Workshop

- Always disconnect devices from power before repair.
- Avoid liquids near electronics.
- Use insulated tools to prevent electric shock.
- Keep the work area clean and organized.
- Use anti-static wristbands when handling circuit boards.

6. Tools for Computer and GSM Repairs

- **Common Tools:** Screwdrivers, multimeter, soldering iron, pliers, brushes, and tweezers.
- **Uses:** Each tool serves a specific purpose—tightening screws, testing power, removing dust, and fixing loose connections.
- Always handle tools properly to prevent damage.

7. Peripheral Devices

- **Meaning:** Devices connected externally to a computer to add functionality.

- **Examples:** Printer, scanner, flash drive, webcam, external hard drive.
- **Importance:** They increase the computer's ability to input, output, and store data.

8. Basic Care and Maintenance of Computer and GSM

- **Meaning:** Keeping devices in good working condition through cleaning, repairs, and careful use.
- **Types:** Preventive and corrective maintenance.
- **Care Practices:** Regular cleaning, antivirus installation, proper charging, and use of original accessories.
- **Importance:** Extends lifespan, improves performance, and prevents data loss.

General Revision Questions

1. Define computer hardware.
2. What is GSM repair?
3. List four input devices.
4. Mention two examples of preventive maintenance.
5. Who is known as the father of the computer?
6. Identify three tools used in GSM repair.
7. Mention two safety rules in the computer workshop.
8. List any three peripheral devices.
9. Explain one importance of maintaining your computer.
10. Mention two career opportunities in computer and GSM repairs.

Practical Revision Activities

1. Students should identify and label computer hardware parts in the ICT lab.
2. Students should practice safe handling of tools like screwdrivers and pliers.
3. Students should demonstrate how to clean a keyboard or phone screen.
4. Teacher should organize a brief class quiz on the topics covered.
5. Group discussion on how computer and GSM repairs can provide income.

Summary

The first term introduced students to the basic concepts of computer hardware and GSM repairs. Students learned about the history, components, tools, and safety measures involved in working with technological devices. Regular care and maintenance were emphasized to ensure durability and efficiency. This revision reinforces these lessons and ensures students are ready for their examinations.

Evaluation

1. Mention three major topics studied during the term.
2. State two safety rules in the workshop.
3. Identify two tools used for computer maintenance.
4. Mention three components of a GSM device.
5. State two reasons why maintenance is important.

Assignment

1. Define the term “maintenance” in relation to computer systems.
2. List five tools used in GSM repairs and their uses.
3. Mention four generations of computers and their characteristics.

4. Explain two functions of peripheral devices.
5. Describe three preventive maintenance practices for a computer system.



JSS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

SECOND TERM

WEEK 1: CLASSIFICATION OF COMPUTERS

Introduction

Computers come in different shapes, sizes, and capacities. They are designed to perform different functions depending on the user's needs. Understanding how computers are classified helps us to know the right type to use for a particular purpose—whether for banking, education, design, or mobile use.

In this lesson, we will learn how computers are classified based on **types** and **sizes**, as well as their different uses in everyday life.

Meaning of Classification of Computers

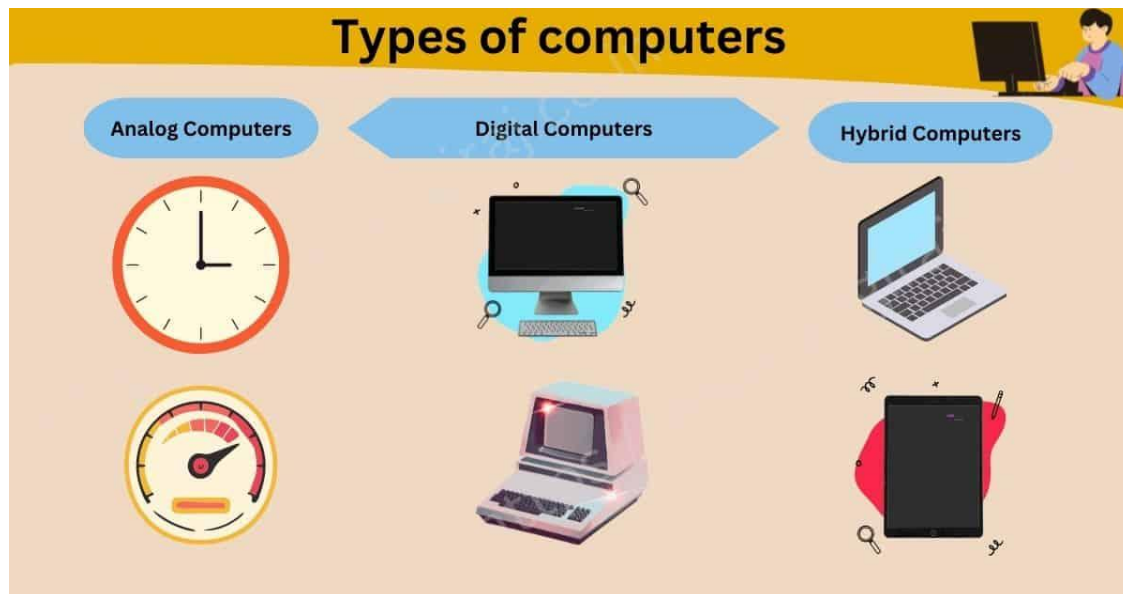
Classification means **grouping computers into different categories** according to their features, processing power, and purpose. Just like we classify animals or plants, computers can also be grouped into different types based on what they do and how powerful they are.

Classification of Computers by Type

There are **three main types of computers** based on how they process data:

1. **Digital Computers**
2. **Analogue Computers**
3. **Hybrid Computers**

Let's examine each one in detail.



1. Digital Computers

Digital computers process data that are in **discrete or numerical form** (i.e., numbers or digits). They are the most common type of computers used today. They operate using binary digits (0s and 1s).

Examples:

Desktop computers, laptops, smartphones, calculators.

Characteristics:

- Process data in numbers or digits.
- Give accurate and fast results.
- Store and retrieve information easily.
- Used in offices, schools, and homes.

2. Analogue Computers

Analogue computers process **continuous data** rather than numbers. They are used to measure physical quantities like speed, temperature, pressure, or voltage.

Examples:

Speedometers in cars, thermometers, voltmeters, and weather monitoring systems.

Characteristics:

- Work with continuously changing data.
- Used mainly for scientific and engineering purposes.
- Less accurate than digital computers but very useful for measuring changes.

3. Hybrid Computers

Hybrid computers combine the features of both **digital** and **analogue** computers. They can process both numerical (digital) and continuous (analogue) data at the same time.

Examples:

Hospital machines like ECG (Electrocardiogram) and flight simulation systems used by pilots.

Characteristics:

- Perform fast calculations (digital) and measure physical data (analogue).
- Used in specialized fields like medicine and research.
- Very expensive and complex to operate.

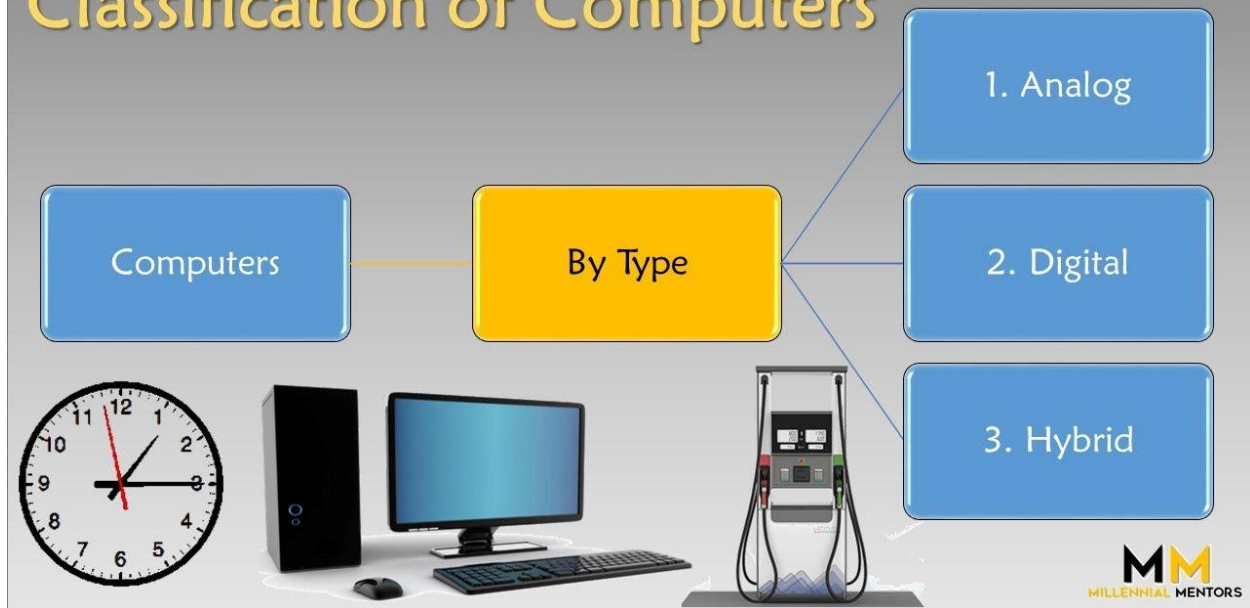
Classification of Computers by Size and Capacity

Computers can also be grouped based on their **size**, **storage capacity**, and **processing power**.

The main classifications are:

1. **Supercomputers**
2. **Mainframe Computers**
3. **Minicomputers**
4. **Microcomputers**

Classification of Computers



1. Supercomputers

These are the **fastest and most powerful** computers. They can process huge amounts of data very quickly and are used for complex scientific work.

Examples: IBM Summit, Cray Titan.

Uses: Weather forecasting, space research, nuclear studies.

2. Mainframe Computers

Mainframe computers are very large systems used by big organizations to process data for many users at once.

Examples: Used by banks, government offices, and airlines.

Features:

- Can process millions of transactions per second.
- Have high storage capacity.
- Support many users at the same time.

3. Minicomputers

These are smaller than mainframes but larger than personal computers. They can serve small organizations and can handle multiple tasks at once.

Examples: Used by small companies and universities for research.

Features:

- Medium processing power.
- Serve as servers in small offices.

4. Microcomputers (Personal Computers)

These are the most common computers used in homes, schools, and offices. They are designed for one user at a time.

Examples: Desktop computers, laptops, tablets, smartphones.

Features:

- Small in size and portable.
- Easy to use and affordable.
- Suitable for typing, gaming, and internet use.

Uses of Different Classes of Computers

Type of Computer	Example	Area of Use
Supercomputer	Cray Titan	Weather and space research
Mainframe	IBM Z Series	Banking and data processing
Minicomputer	PDP-11	University labs

Microcomputer	Laptop, Tablet	Home, school, office
Analogue	Thermometer	Temperature measurement
Hybrid	ECG Machine	Medical diagnosis

Summary

Computers can be classified based on **types** (digital, analogue, and hybrid) and **sizes** (super, mainframe, mini, and micro).

Digital computers handle numbers and symbols, analogue computers measure continuous changes, and hybrid computers combine both features.

Supercomputers and mainframes are used in large organizations, while microcomputers are common in everyday use.

Evaluation

1. What is computer classification?
2. List and explain the three types of computers based on data processing.
3. Mention four classes of computers based on size.
4. Give two examples of analogue computers.
5. State two uses of supercomputers.

Assignment

1. Define the term “computer classification.”
2. Explain the main differences between digital and analogue computers.
3. List four types of computers according to size and explain one use of each.
4. Identify three areas where hybrid computers are used.

5. Draw and label two examples of computer types based on size (micro and mainframe).



WEEK 2: TYPES OF MOBILE PHONES

Introduction

Mobile phones have become essential devices for communication, entertainment, and work. They come in different forms, designs, and capacities. Understanding the types of mobile phones helps users, technicians, and GSM repair students to know their structure, functions, and repair techniques.

Meaning of Mobile Phones

A mobile phone is a wireless electronic device used for communication through voice calls, text messages, and internet access. Modern mobile phones go beyond communication — they now perform several functions like photography, music playback, and computing.

Types of Mobile Phones

Mobile phones can be classified into two major types: **feature phones** and **smartphones**.



1. Feature Phones

Feature phones are basic mobile devices designed mainly for making calls and sending text messages. They have physical keypads and small screens. These phones are durable and have long-lasting batteries.

- **Characteristics:**

- Limited storage capacity
- Small screen display
- No or minimal internet access
- Use of button keypads
- Usually have removable batteries

- **Examples:** Nokia 105, Itel 2160, Tecno T301

- **Advantages:**

- Affordable and easy to use
- Longer battery life
- Durable and less prone to damage

- **Disadvantages:**

- Limited applications and internet capability
- Small display and low camera quality

2. Smartphones

Smartphones are advanced mobile phones that function like mini-computers. They run on operating systems such as Android, iOS, or HarmonyOS, and they can connect to the internet and install applications.

- **Characteristics:**

- Large touchscreen interface
- High-quality camera
- Access to the internet and social media
- Internal and external storage
- GPS, Wi-Fi, and Bluetooth capabilities
- **Examples:** Samsung Galaxy series, iPhones, Tecno Spark, Infinix Note series
- **Advantages:**
 - Multiple functionalities (camera, GPS, internet, etc.)
 - Easy access to online learning and business tools
 - High-quality display and sound
- **Disadvantages:**
 - Expensive to buy and repair
 - Shorter battery life
 - Easily damaged by water or impact

Differences Between Feature Phones and Smartphones

Feature Phones	Smartphones
Physical buttons	Touchscreen interface
Basic communication functions	Multiple advanced functions
Long battery life	Shorter battery life
Small display	Large high-resolution display
No or limited internet access	Full internet access

Practical Activity

Students should bring a feature phone and a smartphone to class. They will identify their parts, compare their functions, and note differences in operation and features.

Evaluation

1. Define a mobile phone.
2. List and explain the two main types of mobile phones.
3. Mention three characteristics of a feature phone.
4. State two differences between a smartphone and a feature phone.
5. Give two examples of smartphones and two examples of feature phones.

Assignment

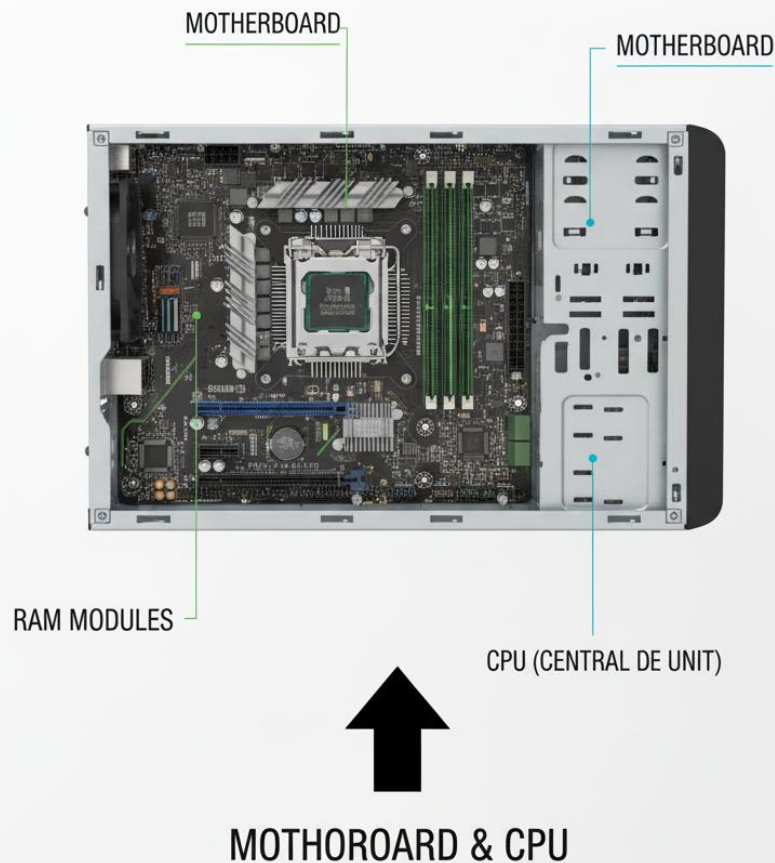
1. Write four advantages of using smartphones.
2. Mention two disadvantages of feature phones.
3. Explain how smartphones have improved communication.
4. List three smartphone operating systems you know.
5. Draw and label the front view of a smartphone.

WEEK 3: SYSTEM UNIT PARTS

Introduction

The system unit is the central part of a computer where most of the essential hardware components are located. It is the heart of the computer system, responsible for processing, storing, and managing all the operations that take place inside the computer. Understanding the parts of the system unit is important for anyone learning computer hardware and GSM repairs because it helps in identifying, assembling, and maintaining the components properly.

INTERNAL VIEW: MOTHERBOARD & CPU



Meaning of System Unit

A system unit is the rectangular box-like case that houses the major internal components of a computer. It is sometimes called the *computer casing* or *CPU box* (though the CPU is just one part inside it). The system unit connects all other computer parts such as the monitor, keyboard, and mouse through cables or wireless connections. It processes data, executes instructions, and stores information.

Main Parts of the System Unit

The system unit contains several key components that work together to ensure the computer performs efficiently. These include:

1. Motherboard

The motherboard is the main circuit board inside the system unit. It serves as the platform on which all other components are connected.

- It houses the CPU, memory (RAM), and expansion slots.
- It allows communication between all parts of the computer.
- It contains connectors for input and output ports.

Example: The motherboard in a desktop computer connects the processor, hard drive, and graphics card to work together.

2. Central Processing Unit (CPU)

The CPU, also known as the *brain of the computer*, carries out all the instructions given to the computer. It performs calculations, logic operations, and controls data movement.

- The CPU has two main parts:
 - a. **Arithmetic and Logic Unit (ALU):** Performs mathematical and logical operations.
 - b. **Control Unit (CU):** Directs and controls all other parts of the computer.
- CPU speed determines how fast a computer can process information.

Examples: Intel Core i5, AMD Ryzen processors.

3. Random Access Memory (RAM)

RAM is the short-term memory of the computer where active programs and data are temporarily stored.

- It is *volatile*, meaning it loses its data when the computer is turned off.
- The more RAM a computer has, the faster it can perform tasks.

Examples: 4GB, 8GB, or 16GB RAM modules used in modern computers.

4. Power Supply Unit (PSU)

The power supply converts electricity from the wall socket (AC) into low-voltage DC power required by the computer.

- It distributes power to the motherboard, drives, and fans.
- Without the PSU, no component in the computer can function.

Examples: ATX and Micro-ATX power supplies.

5. Storage Devices

Storage devices are used to keep data permanently. Unlike RAM, information stored here remains even when the power is turned off.

- **Hard Disk Drive (HDD):** Uses spinning disks to store large amounts of data.
- **Solid State Drive (SSD):** Faster and more durable than HDDs because it has no moving parts.

Examples: 500GB HDD, 256GB SSD.

6. Cooling Fan

The cooling fan helps to reduce the heat generated by the CPU and other components.

- It prevents overheating that can cause computer damage.
- It maintains a stable temperature for optimal performance.

Example: CPU fan, case fan.

7. Ports and Expansion Slots

These are connectors on the motherboard or casing that allow devices like USB drives, printers, and speakers to be attached.

- **Ports:** USB, HDMI, VGA, and Ethernet ports.
- **Expansion Slots:** Allow the addition of cards like sound, graphics, or network cards.

Practical Activity

Students should observe an opened system unit (under teacher supervision). Identify the motherboard, CPU, RAM, power supply, and cooling fan. Discuss how they work together to make the computer function.

Summary

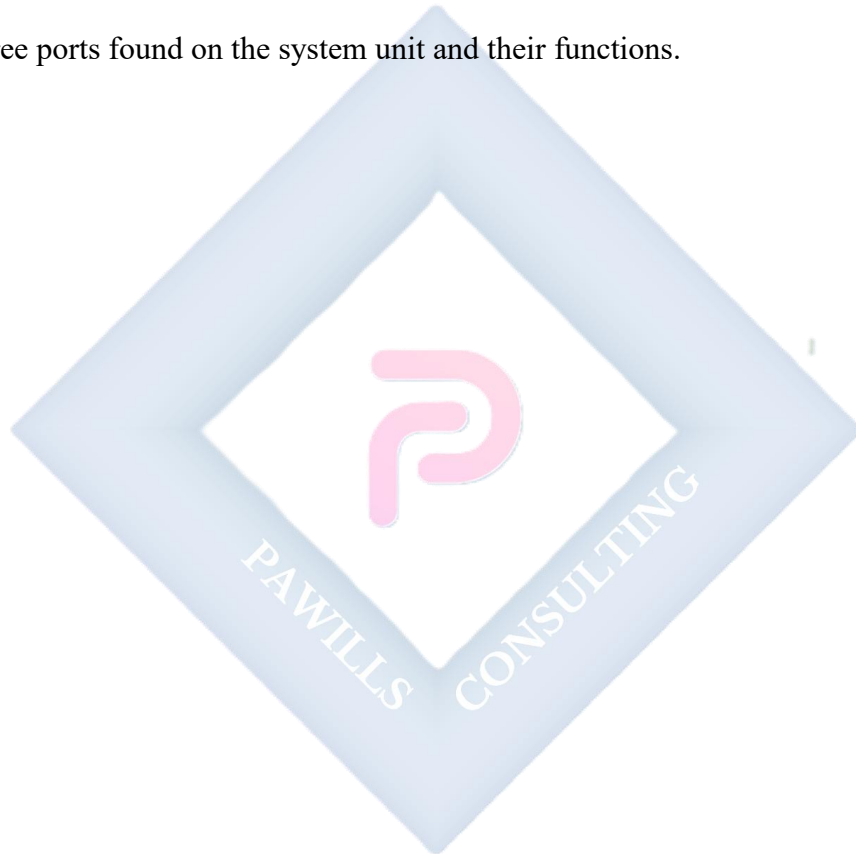
The system unit houses the essential components of a computer that make it work. These include the motherboard, CPU, RAM, power supply, storage devices, cooling fan, and various ports. The CPU processes data, RAM holds temporary information, and the storage devices keep data permanently. All components must work together for the computer to function properly.

Evaluation

1. Define a system unit.
2. Mention four components found inside the system unit.
3. State the functions of the CPU and RAM.
4. Explain the role of the power supply unit.
5. What is the difference between volatile and non-volatile memory?

Assignment

1. Draw and label a diagram of the system unit showing its major components.
2. Explain the function of the motherboard.
3. Describe two differences between HDD and SSD.
4. State what happens when the cooling fan stops working.
5. List three ports found on the system unit and their functions.

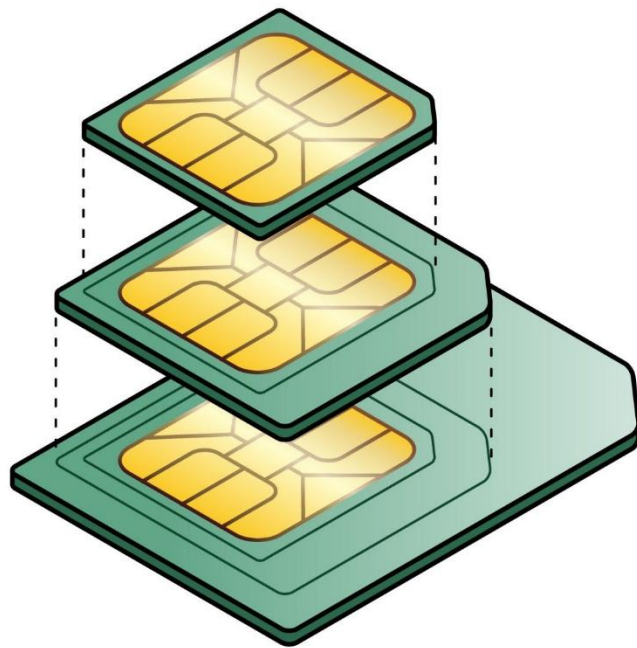


WEEK 4: SIM CARDS AND MEMORY CARDS

Introduction

Modern mobile phones depend on two major removable components — the **SIM card** and the **memory card**. Both play vital roles in the operation and functionality of mobile devices. The SIM card enables communication through mobile networks, while the memory card provides extra storage space for files and applications. For GSM repair technicians, understanding how these cards work, how to handle them safely, and how to troubleshoot related issues is essential.

Nano SIM
Micro SIM
Standard SIM



Meaning of SIM Card

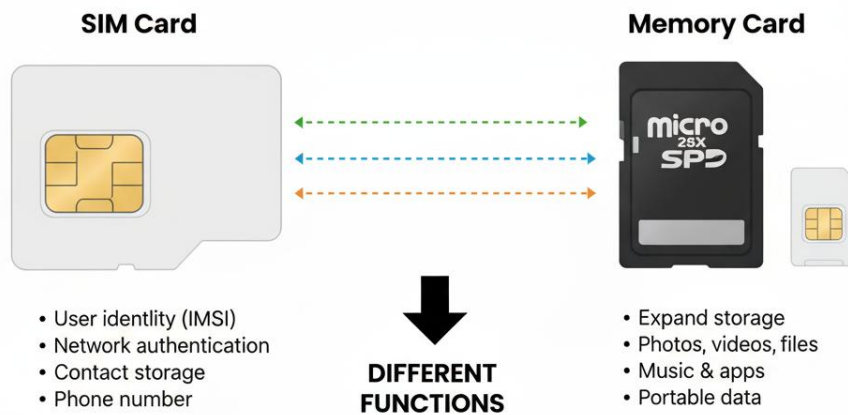
A **SIM card** (Subscriber Identity Module) is a small removable chip inserted into a mobile phone. It contains a unique identification number that connects the phone to a mobile network. The SIM card stores important user information such as the phone number, contact list, network details, and security keys.

Functions of a SIM Card

1. **Network Identification:** It allows the mobile device to connect to a service provider (e.g., MTN, GLO, Airtel, 9mobile).
2. **Storage:** It stores contacts and limited text messages.

3. **Security:** It provides a Personal Identification Number (PIN) and encryption for user data.
4. **Communication Access:** Enables users to make calls, send texts, and use mobile data.
5. **Roaming Support:** Allows users to connect to other networks when traveling.

CARD FUNCTIONS



COMMUNICATION & DATA STORAGE

Types of SIM Cards

SIM cards come in different sizes to fit various mobile devices:

1. **Standard SIM:** The largest type, used in older mobile phones.
2. **Micro SIM:** Smaller in size, used in many mid-generation smartphones.
3. **Nano SIM:** The smallest and most common in modern smartphones.

4. **eSIM (Embedded SIM):** A digital SIM built into the phone that does not require physical removal.

Care and Handling of SIM Cards

- Avoid scratching or bending the metallic chip area.
- Keep away from dust, water, and heat.
- Do not frequently remove or insert the SIM unnecessarily.
- Use the SIM tray properly when inserting into the phone.
- Store unused SIM cards in protective covers.

Meaning of Memory Card

A **memory card** is a small electronic storage device used to store digital data such as photos, music, videos, and applications. Memory cards are widely used in mobile phones, cameras, and other electronic gadgets to expand their storage capacity.

Functions of a Memory Card

1. **Data Storage:** Saves pictures, videos, documents, and music.
2. **Application Storage:** Allows apps to store large files externally.
3. **Data Transfer:** Used to move files from one device to another.
4. **Backup:** Keeps copies of important data in case of phone malfunction.
5. **Performance Enhancement:** Frees up internal memory space for smoother phone operation.

Types of Memory Cards

The most common types of memory cards used in mobile phones and digital devices are:

1. **Secure Digital (SD) Card:** Standard size used in cameras and older phones.
2. **Mini SD Card:** Smaller than SD, now less common.
3. **Micro SD Card:** The smallest and most popular type used in modern smartphones.
4. **SDHC (Secure Digital High Capacity):** Offers larger storage (4GB–32GB).
5. **SDXC (Secure Digital Extended Capacity):** Has very large storage space (above 64GB).

Care and Maintenance of Memory Cards

- Avoid exposure to moisture, dust, and extreme temperatures.
- Eject safely before removing from the device.
- Do not bend or force into the card slot.
- Keep backup copies of important files.
- Format the card only when necessary and use compatible devices.

Differences Between SIM Cards and Memory Cards

Features	SIM Card	Memory Card
Main Function	Connects to mobile network	Stores files and applications
Data Stored	Phone number, contacts, text messages	Photos, videos, documents
Type of Device	Telecom device	Storage device
Dependence	Operates under mobile network	Works independently
Replacement	Requires activation	Can be used instantly

Common Problems and Solutions

1. **SIM Not Detected:** Clean the SIM's metal area and reinsert properly.
2. **Memory Card Error:** Format or check compatibility.
3. **Data Corruption:** Use antivirus and avoid unsafe removal.
4. **Card Locked:** Enter correct PIN or password to unlock.
5. **Slow Reading:** Use high-speed memory cards for better performance.

Practical Activity

Students should observe and handle different types of SIM cards and memory cards. They will insert and remove them safely from demo phones under the teacher's supervision, noting the size and fit.

Summary

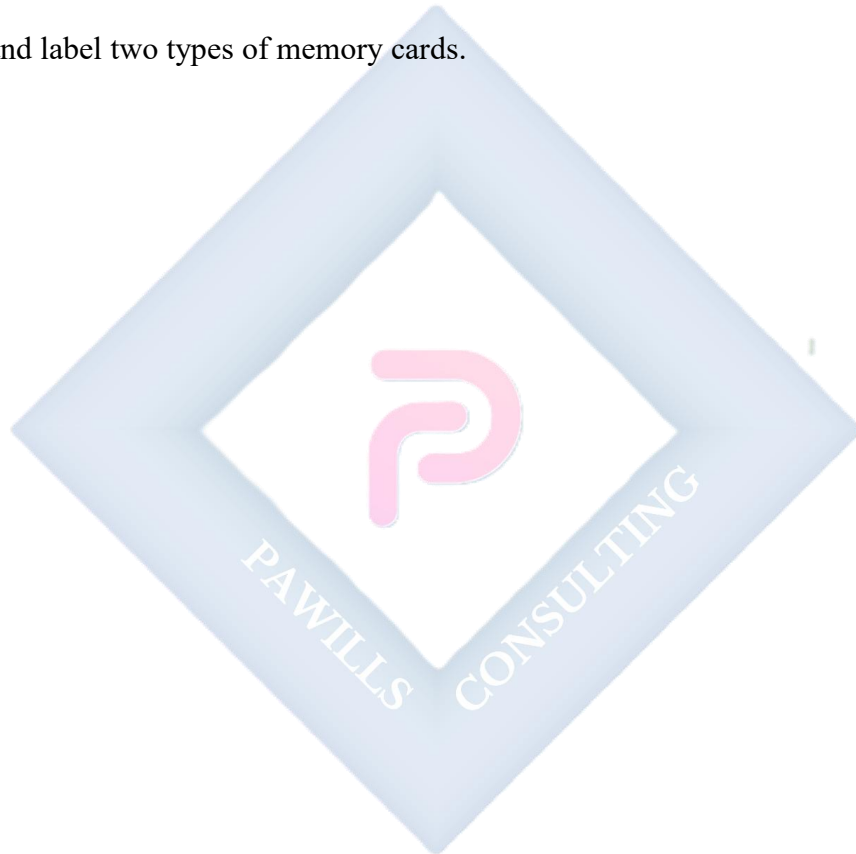
SIM cards and memory cards are essential removable components of modern phones. The SIM card connects users to mobile networks, while the memory card provides storage for digital content. Proper care, handling, and understanding of their functions are important for users and repair technicians.

Evaluation

1. Define a SIM card.
2. List three functions of a SIM card.
3. Mention two types of memory cards used in mobile phones.
4. State three precautions for handling a memory card.
5. Differentiate between a SIM card and a memory card.

Assignment

1. Explain the meaning of SIM and memory cards.
2. List four types of SIM cards and their sizes.
3. Describe three uses of memory cards in mobile phones.
4. State two problems that may affect SIM cards and how to fix them.
5. Draw and label two types of memory cards.



WEEK 6: INTRODUCTION TO SOFTWARE

Introduction

In the world of computers and mobile devices, software plays a vital role in making the hardware function effectively. While hardware refers to the physical components of a computer or phone, software is the set of instructions or programs that tell the hardware what to do. Without software, a computer or GSM device cannot perform any meaningful task. Therefore, understanding software is essential for anyone studying computer hardware and GSM repairs.

Meaning of Software

Software can be defined as a collection of data, programs, or instructions that enable a computer or mobile phone to perform specific operations. It is the non-physical part of a computing system that directs the hardware. Software makes it possible for users to type documents, browse the internet, play music, and communicate through phones.

In simple terms, software is what gives life to hardware. For instance, the same phone hardware can act as a calculator, camera, or music player depending on the software installed on it.

Characteristics of Software

1. **Intangible:** Software cannot be touched or held like hardware components.
2. **Easily Modifiable:** Software can be updated or changed to improve performance.
3. **Dependent on Hardware:** It cannot work alone; it needs hardware to run.
4. **Error-Prone:** Software can contain bugs or errors that affect performance.
5. **Created by Programming:** It is written using programming languages by developers.

Types of Software

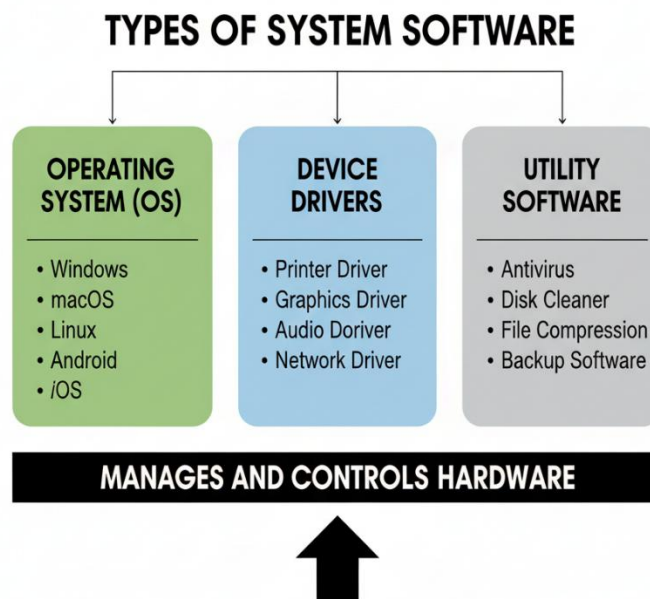
Software can be divided into two main categories: **System Software** and **Application Software**.

1. System Software

System software is the basic software that manages and controls the operations of a computer or phone. It serves as a bridge between the user and the hardware. Without system software, the computer or mobile device cannot start, recognize hardware, or run other applications.

Examples of System Software:

- **Operating Systems:** Windows, Android, iOS, macOS, Linux.
- **Utility Programs:** Antivirus, file manager, disk cleanup, battery saver.
- **Device Drivers:** Software that allows hardware components such as printers, keyboards, or cameras to function properly.
- **Firmware:** Software embedded into hardware components like mobile phone chips or computer BIOS.



Functions of System Software:

1. Controls and manages hardware components.
2. Provides an environment for running applications.
3. Handles memory allocation and storage.
4. Ensures security through user authentication.
5. Facilitates communication between software and hardware.

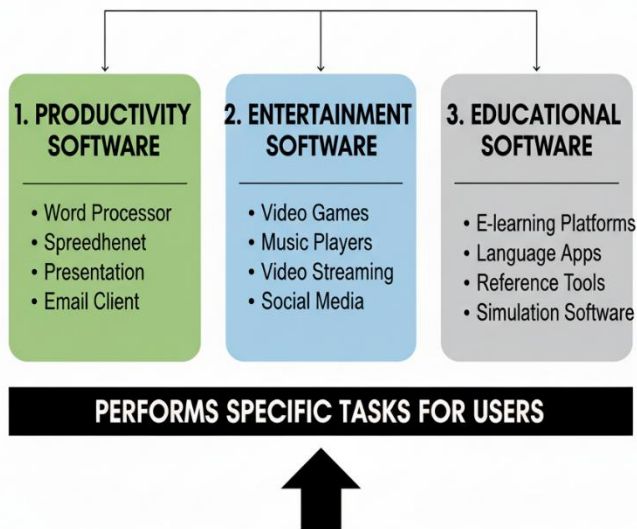
2. Application Software

Application software refers to the programs designed to perform specific tasks for the user. These are the tools that help people carry out practical activities on their computers or phones.

Examples of Application Software:

- **Word Processing Programs:** Microsoft Word, Google Docs, WPS Office.
- **Spreadsheet Programs:** Microsoft Excel, Google Sheets.
- **Graphics and Design Software:** CorelDRAW, Photoshop, Canva.
- **Communication Apps:** WhatsApp, Telegram, Zoom, Messenger.
- **Media Players:** VLC, Windows Media Player.
- **Games:** Candy Crush, Chess, FIFA, etc.

TYPES OF APPLICATION SOFTWARE



Functions of Application Software:

1. Helps users perform specialized tasks such as writing, designing, or communicating.
2. Improves productivity in offices, schools, and homes.
3. Provides entertainment through games, music, and videos.
4. Enhances communication via emails, chats, and video calls.
5. Aids in learning and education through tutorials and e-books.

Differences Between System Software and Application Software

Features	System Software	Application Software
Purpose	Controls and manages hardware	Performs specific user tasks
Operation	Runs automatically when the system starts	Runs when the user opens it

Examples	Windows, Android, iOS	MS Word, WhatsApp, VLC
Dependency	Works before application software	Depends on system software
Installation	Usually pre-installed	Installed by the user

Importance of Software in Computers and Phones

1. It enables communication between users and devices.
2. It allows hardware components to function correctly.
3. It helps perform various operations such as typing, browsing, and designing.
4. It improves the efficiency and performance of devices.
5. It supports troubleshooting and repairs through diagnostic programs.

Common Problems Related to Software

1. **Software Crash:** When a program stops working due to errors.
2. **Virus Attack:** When malicious software affects performance.
3. **Incompatibility:** When software cannot run on a device due to hardware or system differences.
4. **Slow Performance:** Caused by too many background programs.
5. **Failed Updates:** When software does not install properly.

Solutions to Software Problems

1. Install reliable antivirus software to protect against malware.
2. Keep software regularly updated.
3. Remove unused applications to free up space.

4. Restart the device to clear temporary errors.
5. Reinstall or reset corrupted programs.

Summary

Software is a vital part of every computer and mobile device. It gives instructions to hardware and enables users to perform tasks. There are two major types: **System software**, which controls the overall operation of the device, and **Application software**, which allows users to carry out specific functions. Understanding both types helps in diagnosing and repairing computer or phone issues effectively.

Evaluation

1. What is software?
2. List and explain the two main types of software.
3. Give three examples each of system software and application software.
4. Mention three differences between system and application software.
5. State two importance of software in a computer or GSM phone.

Assignment

1. Define software in your own words.
2. Explain the functions of an operating system.
3. Identify five application software and their uses.
4. State three causes of software failure in phones.
5. Discuss two ways to maintain good software performance on a mobile device.

WEEK 8: POWER SUPPLY IN COMPUTERS AND PHONES

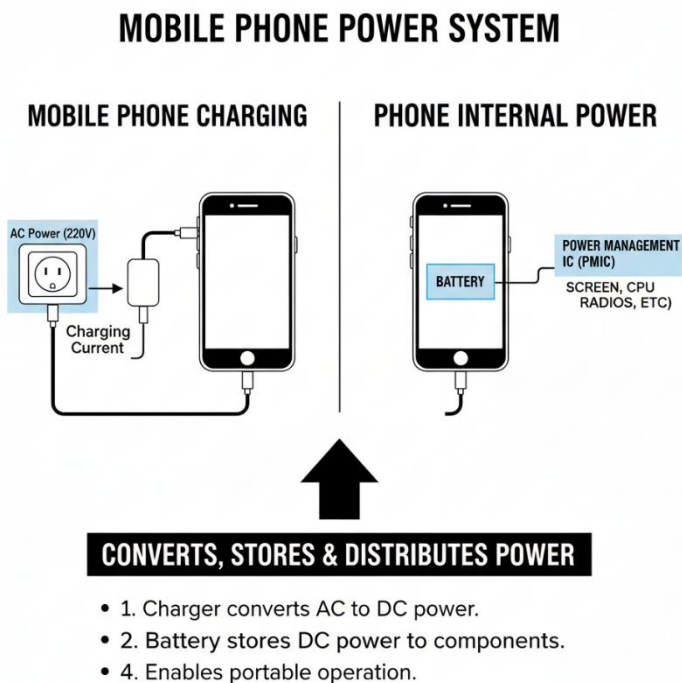
Introduction

Every electronic device — whether a computer or a mobile phone — needs power to function. Power supply is the process of providing electrical energy to components that make a device work. Without proper power supply, computers will not boot, and mobile phones will not charge or operate. In this lesson, we will study how computers and phones receive power, the components involved, types of power supplies, and how to maintain them.

Meaning of Power Supply

A power supply refers to the source or system that provides electrical power to operate devices. It converts electrical energy from one form (like AC from the wall socket) into another form (like DC used by computer or phone components).

In simpler terms, it ensures that every part of the computer or mobile device receives the correct voltage and current needed for it to perform effectively.



Power Supply in Computers

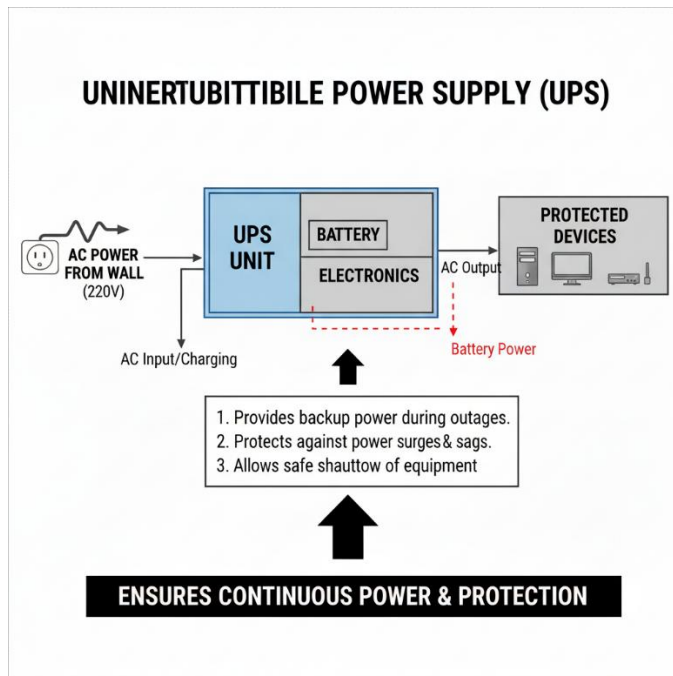
Computers use a **Power Supply Unit (PSU)** to convert Alternating Current (AC) from the wall outlet into Direct Current (DC) needed by internal components. The PSU distributes the required voltage to the motherboard, hard drive, CD-ROM, and other peripherals.

Main Components of a Computer Power System:

1. **Power Cable:** Connects the computer to the electric socket.
2. **Power Supply Unit (PSU):** Converts AC to DC power.
3. **Motherboard Power Connectors:** Supply power from the PSU to the motherboard.
4. **Cooling Fan:** Prevents the power unit and other parts from overheating.
5. **Battery (CMOS):** Maintains system settings such as date and time when the system is off.
6. **Uninterruptible Power Supply (UPS):** Provides backup power when there is an outage, preventing data loss.

Types of Computer Power Supply

1. **Linear Power Supply:** Provides a stable output voltage but wastes energy as heat. Used in older systems.
2. **Switch Mode Power Supply (SMPS):** Converts power efficiently with minimal heat. Common in modern computers.
3. **Uninterruptible Power Supply (UPS):** Offers temporary power during an outage and protects the system from damage.



4. **Power Adapter:** Used for laptops; converts AC to DC to charge the battery and power the system.

Functions of Computer Power Supply

- Supplies stable voltage to system components.
- Converts AC to DC current.
- Protects the computer from power fluctuations.
- Prevents overheating and damage to electrical parts.
- Ensures data safety by maintaining continuous power flow through UPS.

Power Supply in GSM Phones

Mobile phones rely on **batteries and chargers** for their power supply. Unlike computers, they are portable and therefore use rechargeable batteries.

Main Components of a GSM Power System:

1. **Battery:** Stores and supplies electrical energy to the phone's circuits.
2. **Charger:** Converts AC power to DC suitable for charging the battery.
3. **Charging Port and Cable:** Transfers current from the charger to the battery.
4. **Power Management Integrated Circuit (PMIC):** Regulates voltage distribution to the phone's components.
5. **Charging IC (Integrated Circuit):** Controls the charging process to prevent overcharging.

Types of Phone Batteries

1. **Nickel-Cadmium (Ni-Cd):** Older type, heavy, and suffers from memory effect.
2. **Nickel-Metal Hydride (Ni-MH):** Improved capacity but still less efficient than modern batteries.
3. **Lithium-Ion (Li-ion):** Lightweight, durable, and holds charge longer. Common in most phones.
4. **Lithium-Polymer (Li-Po):** Advanced version of Li-ion, used in high-end smartphones.

Functions of GSM Phone Power Components

- **Battery:** Provides the energy needed for phone operations.
- **Charger:** Supplies power to charge the battery.
- **PMIC and Charging IC:** Manage voltage and prevent overcharging or overheating.
- **Charging Port:** Connects the phone to external power.

Importance of Power Supply in Computers and Phones

1. Ensures smooth functioning of devices.

2. Prevents data loss and damage from sudden power cuts.
3. Protects internal circuits from power surges.
4. Maintains device performance and stability.
5. Enhances battery lifespan through regulated charging.

Care and Maintenance of Power Supply Units and Batteries

1. Always use the correct voltage rating for devices.
2. Avoid exposing power units and chargers to water or heat.
3. Use certified and original chargers and adapters.
4. Unplug computers and chargers during lightning storms.
5. Clean dust from PSU fans regularly.
6. Avoid overcharging mobile phone batteries.
7. Replace faulty cables and power cords immediately.
8. Use surge protectors or UPS for computers.

Signs of Faulty Power Supply or Battery

- Computer or phone not powering on.
- Random shutdowns or restarts.
- Overheating during operation or charging.
- Unusual sounds from PSU fan.
- Swollen or leaking batteries.

Troubleshooting Power Supply Problems

1. If a computer doesn't start:

- Check the power cable and socket.
- Test PSU with a power tester.
- Check for loose motherboard connectors.

2. If a phone doesn't charge:

- Inspect the charger, charging cable, and port.
- Try another charger to confirm fault.
- Replace the battery if swollen or not holding charge.

Summary

Power supply provides the energy needed for computers and phones to operate. Computers rely on PSUs, UPS units, and adapters, while mobile phones depend on batteries and chargers. Proper care and maintenance prevent electrical faults and prolong device lifespan. Understanding power systems also helps GSM and computer repair technicians to detect and fix power-related problems efficiently.

Evaluation

1. What is a power supply?
2. List four components of a computer power system.
3. Mention two functions of a UPS.
4. State three types of phone batteries.
5. Identify two signs of a faulty power supply.

Assignment

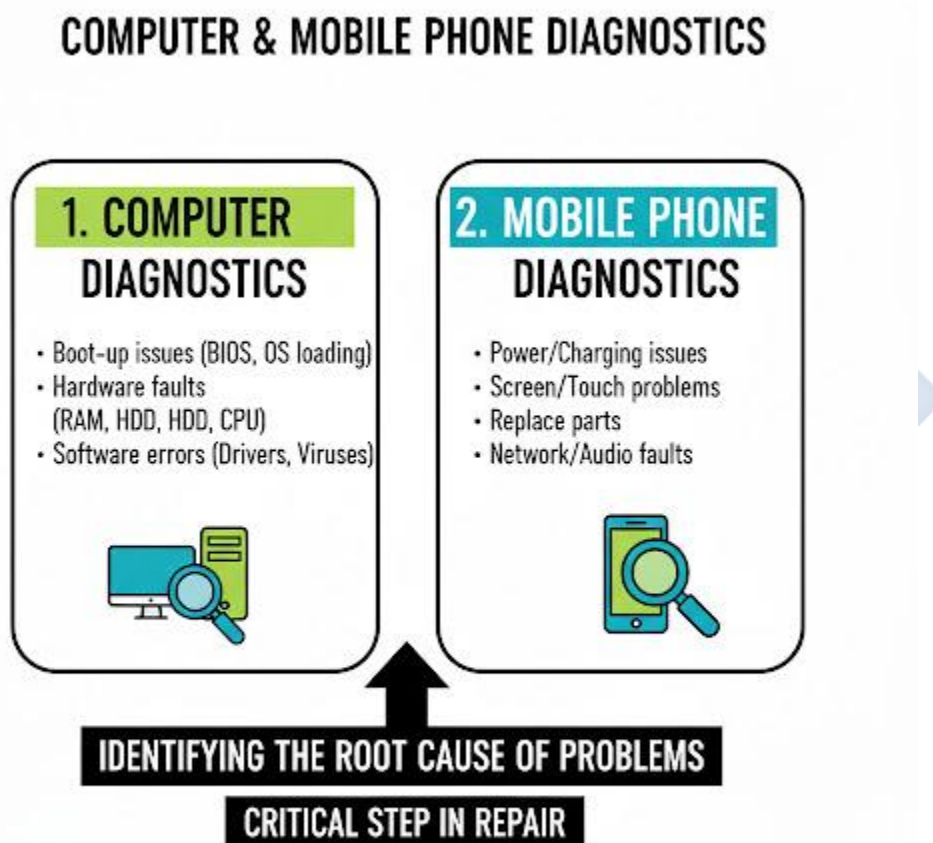
1. Explain how power is supplied to a desktop computer.
2. Differentiate between a laptop power adapter and a GSM charger.
3. Mention three ways of maintaining a phone's battery.
4. List four functions of the Power Supply Unit (PSU).
5. Describe two troubleshooting steps for a computer that won't power on.



WEEK 9: BASIC TROUBLESHOOTING

Introduction

Every computer and mobile phone user will, at some point, experience a fault or malfunction. The process of identifying, analyzing, and fixing these faults is called **troubleshooting**. In this lesson, we will explore the meaning of troubleshooting, the steps involved, common problems in computers and GSM phones, and how to solve them. Understanding basic troubleshooting helps users and technicians save time, reduce repair costs, and keep devices in good working condition.



Meaning of Troubleshooting

Troubleshooting is the process of finding the cause of a problem or fault in a system and fixing it. It involves careful observation, testing, and replacing or repairing defective components. In computer and GSM repairs, troubleshooting means using logical steps to discover why a system or device is not working properly.

For example:

- When a computer refuses to boot, a technician checks the power cable, monitor connection, and system unit.
- When a phone refuses to charge, the technician checks the charger, cable, and battery.

Importance of Troubleshooting

1. Helps identify and fix faults quickly.
2. Reduces repair and replacement costs.
3. Saves time by avoiding unnecessary part changes.
4. Increases the lifespan of computers and phones.
5. Improves maintenance skills of users and technicians.
6. Ensures continuous and efficient operation of devices.

Steps in Troubleshooting

To effectively solve any computer or phone problem, the following steps are used:

1. **Observation:** Carefully observe the device for any unusual behavior (e.g., no display, noise, or overheating).
2. **Identification of the Problem:** Determine what exactly is wrong — e.g., no power, no display, or battery not charging.
3. **Possible Cause:** Think of what might have caused the problem — power failure, software issue, loose cable, etc.
4. **Testing and Diagnosis:** Use tools such as multimeters, power testers, or inspection methods to check for faults.
5. **Repair or Replacement:** Fix or replace faulty components after confirmation.

6. **Verification:** Test the device again to ensure the problem is solved.
7. **Documentation:** Record what was done, tools used, and the final solution for future reference.

Common Computer Problems and Solutions

Problem	Possible Cause	Solution
Computer not booting	Faulty PSU, loose cables, bad motherboard	Check and tighten cables, test PSU, replace defective parts
No display on monitor	Loose VGA/HDMI cable, faulty monitor	Reconnect or replace monitor cable
Keyboard or mouse not working	Faulty port or device	Try another device or replace the port
Computer overheating	Dust in fan, poor ventilation	Clean cooling fan and ensure airflow
Slow performance	Too many applications, virus infection	Uninstall unnecessary programs, run antivirus scan

Common GSM Phone Problems and Solutions

Problem	Possible Cause	Solution
Phone not charging	Faulty charger, bad charging port, weak battery	Test with another charger, clean or replace charging port
Phone not powering on	Battery problem, power IC fault	Charge or replace battery, check power IC
No network signal	Damaged antenna, SIM issue	Check SIM card, replace antenna if faulty

Screen not displaying	Broken LCD or loose connector	Replace or reconnect display screen
Phone freezing or hanging	Too many apps, low memory	Restart phone, delete unused apps, clear cache

Troubleshooting Tools and Equipment

To carry out effective troubleshooting, technicians use certain basic tools:

1. **Multimeter:** For checking voltage, resistance, and continuity.
2. **Screwdrivers:** For opening system units and phone cases.
3. **Power Tester:** For testing PSUs or power outlets.
4. **Soldering Iron:** For fixing damaged circuits or connectors.
5. **Brush and Blower:** For cleaning dust and debris from devices.
6. **Magnifying Glass:** For inspecting tiny components on circuit boards.

Safety Precautions During Troubleshooting

1. Always switch off and unplug devices before opening them.
2. Avoid working on wet surfaces or with wet hands.
3. Use insulated tools to prevent electric shock.
4. Handle circuit boards carefully to avoid static electricity damage.
5. Do not test high-voltage circuits without proper knowledge.
6. Keep metallic objects away from live components.

Difference Between Computer and GSM Troubleshooting

Computer Troubleshooting	GSM Troubleshooting
Involves larger components like PSU, motherboard, and peripherals.	Involves smaller electronic parts like charging port, battery, and IC.
Requires desktop tools such as power tester and UPS.	Uses precision tools such as soldering iron and tweezers.
Common faults: boot failure, overheating, and display issues.	Common faults: charging issues, screen failure, network loss.
Operates mainly on AC power.	Operates mainly on DC battery power.

Summary

Troubleshooting is the process of detecting and fixing faults in computers and mobile phones. It involves observation, diagnosis, testing, and repair. Knowing how to troubleshoot helps technicians maintain devices, prevent damage, and save money. Both computer and GSM troubleshooting require patience, skill, and the use of appropriate tools.

Evaluation

1. Define troubleshooting.
2. State four steps involved in troubleshooting a faulty system.
3. Mention three common GSM phone problems and their possible causes.
4. List four tools used for troubleshooting.
5. Explain two safety precautions to observe during troubleshooting.

Assignment

1. Explain in detail the process of troubleshooting a computer that does not boot.

2. Identify five common GSM phone faults and suggest possible solutions.
3. List and describe four troubleshooting tools and their uses.
4. Mention three differences between computer and GSM troubleshooting.
5. State four reasons why troubleshooting is important in computer and GSM maintenance.

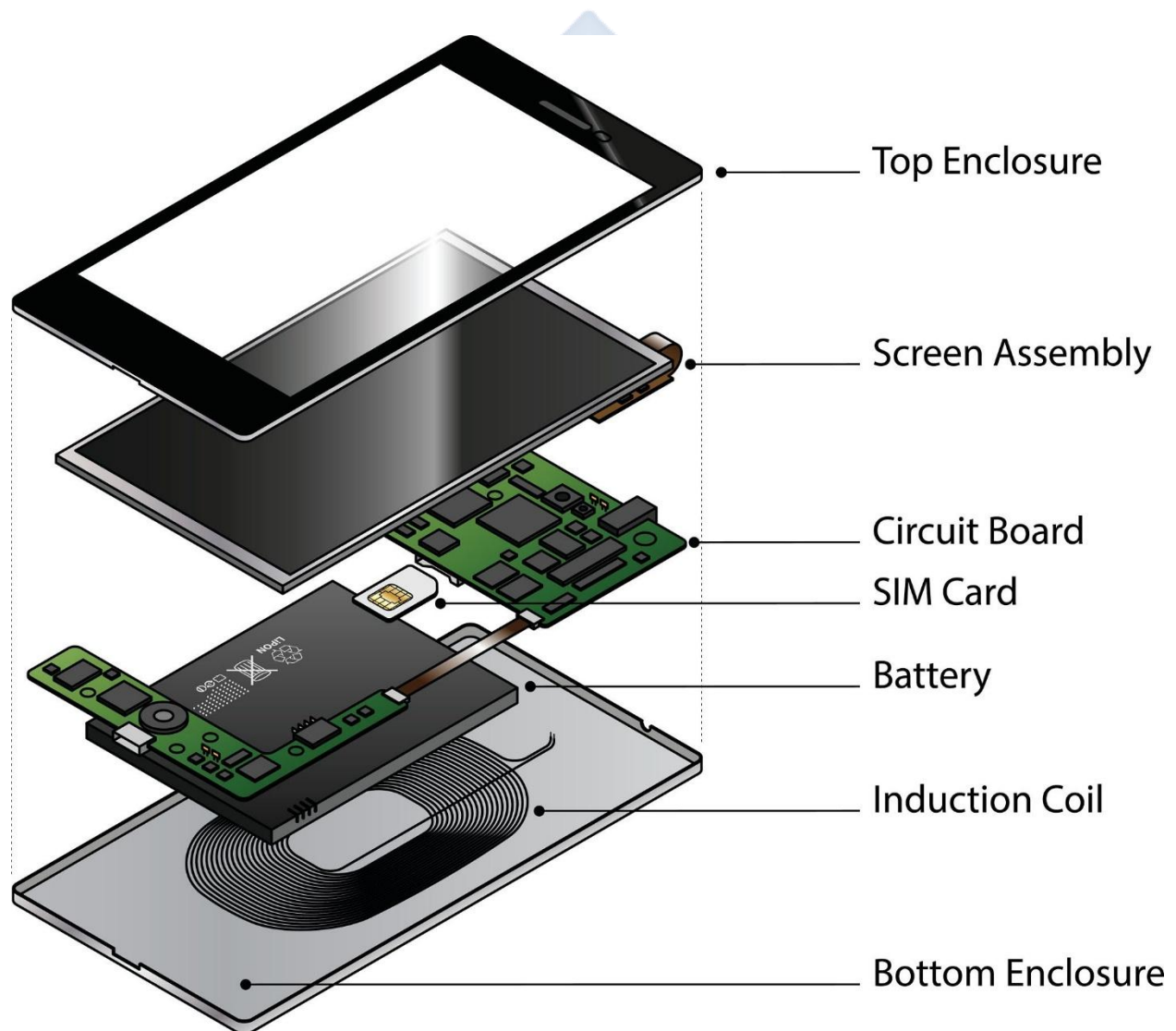


WEEK 10: PRACTICAL – DISASSEMBLING A PHONE

Introduction

In mobile phone repairs, understanding how to **disassemble (open)** and **reassemble (close)** a phone safely is one of the most important practical skills. Disassembling a phone means carefully separating its external and internal components to identify faults, clean parts, or replace damaged components.

This lesson will guide students through the basic steps, tools, and precautions needed to safely open a phone, identify its major parts, and put it back together correctly.



Meaning of Disassembling a Phone

Disassembling a phone is the process of opening a mobile device to inspect, repair, or replace

internal components such as the screen, battery, charging port, speaker, or circuit board. It must be done with **care and the right tools** to avoid damaging delicate parts.

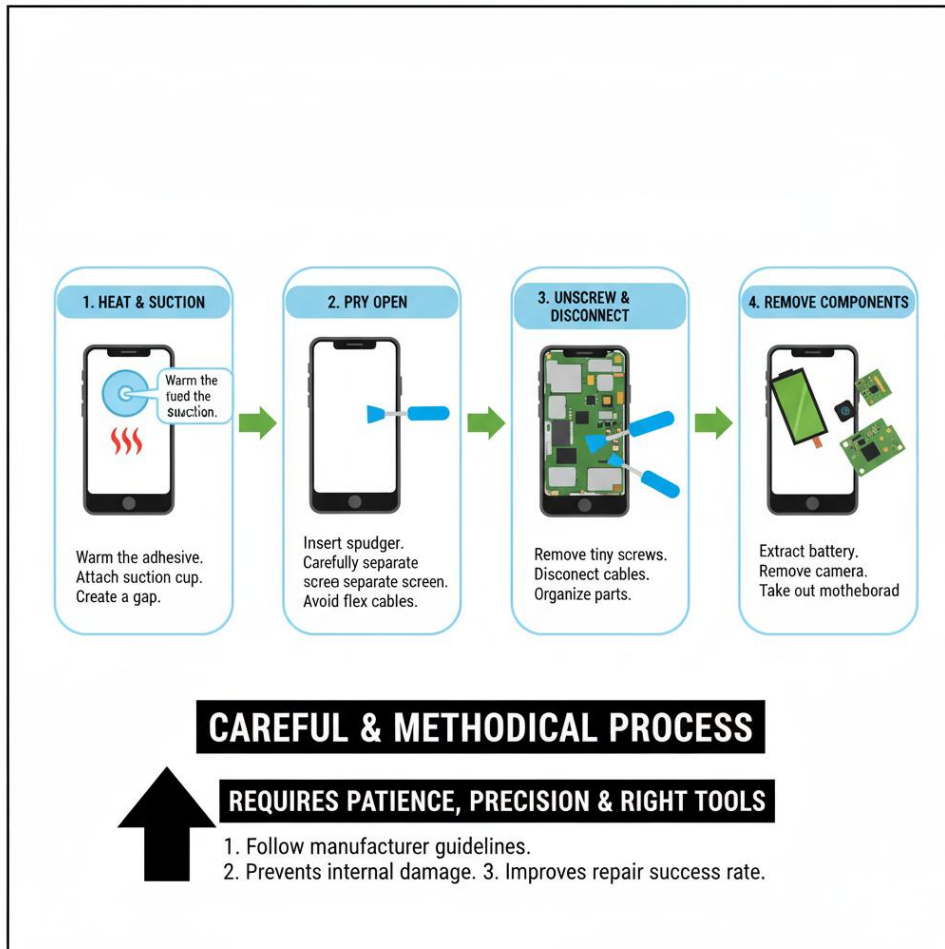
Importance of Learning to Disassemble a Phone

1. Helps in identifying and repairing internal faults.
2. Enables technicians to clean or replace damaged parts.
3. Improves understanding of phone structure and design.
4. Develops students' practical and technical skills.
5. Helps in recycling or salvaging useful phone parts.
6. Prevents total phone damage during repair.

Tools Required for Disassembling a Phone

To safely open and repair a phone, the following tools are commonly used:

1. **Small Screwdrivers (Phillips and Flat Head):** Used to remove or tighten screws.
2. **Plastic Pry Tools:** Used to open phone casings without scratching them.
3. **Tweezers:** Used to pick or handle small components.
4. **Suction Cup:** Used to remove screens safely without cracking them.
5. **Soldering Iron and Solder Wire:** Used to remove or attach components on the circuit board.
6. **Anti-static Wrist Strap:** Prevents electrostatic discharge that can damage parts.
7. **Magnifying Glass or Lamp:** Helps to see tiny parts clearly.
8. **Brush and Blower:** Used to remove dust or dirt from phone components.
9. **SIM Card Ejector Tool:** Used to remove SIM and memory cards.



10.

Main Components of a Mobile Phone

When a phone is disassembled, the following major components can be found:

1. **Casing (Housing):** The outer body that protects the phone's internal parts.
2. **Battery:** Supplies power to the phone.
3. **Screen (LCD/Touchscreen):** Displays images and allows user interaction.
4. **Motherboard (Logic Board):** The main circuit that connects all components.
5. **Camera Module:** Captures photos and videos.
6. **Speaker and Microphone:** Used for sound input and output.

7. **Charging Port:** Connects the charger to the phone.
8. **Antenna:** Enables network connectivity.
9. **SIM and Memory Card Slot:** Holds SIM and SD cards.
10. **Power and Volume Buttons:** Controls functions like power and sound.

Step-by-Step Process of Disassembling a Phone

1. Preparation:

- Switch off the phone completely.
- Remove SIM and memory cards.
- Gather all necessary tools and work on a clean surface.

2. Remove the Back Cover and Battery:

- Open the back cover using a pry tool or fingernail.
- Carefully remove the battery if it's removable.

3. Unscrew and Separate the Casing:

- Use the right screwdriver to remove all visible screws.
- Keep screws in a safe place to avoid losing them.

4. Disconnect Internal Components:

- Use tweezers to gently remove ribbon connectors and other parts.
- Disconnect the display, camera, and charging port connections.

5. Remove the Motherboard:

- Gently lift the motherboard out of the casing.
- Be cautious not to damage any attached cables or chips.

6. Inspect and Identify Faulty Parts:

- Examine the board and other parts for burnt, swollen, or broken components.

7. Reassemble the Phone:

- Replace all parts in reverse order, ensuring every connection fits properly.
- Tighten screws and close the casing securely.
- Test the phone to confirm it powers on and functions well.

Safety Precautions During Phone Disassembly

1. Always switch off the phone before disassembly.
2. Avoid using excessive force when opening parts.
3. Use only the correct tools for each task.
4. Keep screws and small parts in labeled containers.
5. Wear anti-static gloves or wrist straps to prevent electrical damage.
6. Do not touch components with wet or oily hands.
7. Avoid short-circuiting parts with metal tools.
8. Handle the screen and motherboard carefully.

Common Mistakes to Avoid

1. Using the wrong screwdriver type or size.
2. Forcing components apart without unlocking connectors.
3. Misplacing or mixing up screws.
4. Forgetting to disconnect the battery before soldering.
5. Reassembling without testing each part.

Benefits of Proper Disassembly Practice

- Prevents damage to phone parts.
- Makes repairs faster and easier.
- Helps in proper cleaning and maintenance.
- Builds confidence for handling advanced repairs.
- Encourages organization and attention to detail.

Summary

Disassembling a phone involves carefully opening the device, identifying its internal components, and reassembling it correctly after repair or inspection. The process requires precision, patience, and the right tools. Proper disassembly skills are essential for every aspiring GSM technician and help maintain device safety and performance.

Evaluation

1. Define phone disassembly.
2. List five tools used in disassembling a phone.
3. Mention four internal parts of a mobile phone.
4. State three safety precautions during phone disassembly.
5. Explain why it is important to disconnect the battery before repair.

Assignment

1. Write out seven steps involved in disassembling a phone.
2. Mention five mistakes that should be avoided during disassembly.
3. List ten components of a phone you may find when it is opened.

4. Explain two reasons why a technician must reassemble the phone carefully.
5. Describe the function of the motherboard and charging port in a phone.



WEEK 11: REVISION – REVIEW OF KEY TOPICS

Introduction

The revision week gives students the opportunity to recall, review, and consolidate all that has been taught during the second term in *Computer Hardware and GSM Repairs*. The aim of this session is to strengthen understanding, correct misconceptions, and ensure mastery of key topics before examinations.

Summary of Major Topics Covered in the Term

1. Classification of Computers

Computers are classified based on *types* and *sizes*.

- **By Type:** Digital, Analogue, and Hybrid computers.
 - *Digital computers* process data in discrete form (e.g., PCs, laptops).
 - *Analogue computers* measure continuous data (e.g., speedometers).
 - *Hybrid computers* combine both features (used in hospitals and industries).
- **By Size and Capacity:**
 - *Microcomputers* – small, personal devices.
 - *Minicomputers* – medium-sized, multi-user systems.
 - *Mainframes* – large computers for many users.
 - *Supercomputers* – fastest and most powerful systems.

2. Types of Mobile Phones

- **Feature Phones:** Basic phones with calling and texting functions, long battery life, and limited apps.
- **Smartphones:** Advanced phones with internet connectivity, touchscreens, and downloadable apps.
- Differences include operating system, storage capacity, and functionality.

3. System Unit Parts

The *system unit* houses major components such as:

- **Motherboard:** The main circuit board connecting all parts.
- **Central Processing Unit (CPU):** The “brain” of the computer.
- **Random Access Memory (RAM):** Temporary storage for active tasks.
- **Power Supply Unit (PSU):** Converts electrical power to usable form.
- **Hard Disk Drive (HDD):** Permanent data storage.

4. SIM and Memory Cards

- **SIM Card:** Subscriber Identity Module; stores network information and user data.
- **Memory Card:** External storage for photos, videos, and apps.
- Proper care includes keeping them dry, avoiding scratches, and ejecting safely.

5. Introduction to Software

- **System Software:** Runs the hardware (e.g., operating system like Windows or Android).
- **Application Software:** Performs user-specific tasks (e.g., MS Word, WhatsApp).
- Both work together to make computers and phones functional.

6. Power Supply in Computers and Phones

- **Batteries:** Provide power for portable devices.
- **Chargers and Power Adapters:** Convert AC to DC current for charging.
- **Uninterruptible Power Supply (UPS):** Protects computers from power surges or outages.
- Overcharging and using fake chargers can damage devices.

7. Basic Troubleshooting

- Troubleshooting means identifying and solving technical faults.
- Common issues:
 - Computer not booting → check power cable or battery.
 - Phone not charging → check charger or charging port.
 - Device overheating → close unused apps, ensure ventilation.
- Steps include *observation*, *diagnosis*, *testing*, and *correction*.

8. Practical: Disassembling a Phone

- Steps: Power off → Remove battery → Unscrew → Open casing → Identify parts → Test → Reassemble.
- Tools: Screwdrivers, pry tools, tweezers, anti-static strap.
- Precautions: Avoid static electricity, handle gently, and keep parts organized.

Practical Revision Questions

1. Differentiate between a *feature phone* and a *smartphone*.
2. State three differences between *system software* and *application software*.
3. List four internal components of the *system unit*.
4. What is a *SIM card* and what is its function?
5. Mention three safety precautions when disassembling a phone.
6. Define *troubleshooting* and give one example each for computer and phone faults.
7. Why is the motherboard important in both phones and computers?
8. List any three power devices used in phones or computers.
9. Explain the term *analogue computer* and give one example.
10. Describe three steps involved in opening a phone safely.

Summary of the Term's Learning Objectives

By the end of the term, students should be able to:

- Identify and classify computers by type and size.
- Distinguish between different types of mobile phones.
- Identify the main parts of the system unit and their functions.
- Explain the uses and care of SIM and memory cards.
- Differentiate between system and application software.
- Understand power sources and their maintenance.
- Diagnose and solve simple hardware and phone issues.
- Safely disassemble and reassemble a mobile phone.

Evaluation

1. Explain what hybrid computers are and their uses.
2. Describe the function of the CPU and RAM.
3. Differentiate between feature phones and smartphones.
4. State two reasons for a phone not charging.
5. List the steps for reassembling a phone after repair.

Assignment

1. Prepare a summary chart showing all topics learned this term.
2. Create a short note on the precautions to observe in a repair workshop.
3. List ten tools used in GSM and computer repair with their uses.

4. Explain three benefits of learning computer hardware and GSM repairs.
5. Write five possible examination questions from any topic this term and provide answers.



JSS1 COMPUTER HARDWARE AND GSM REPAIRS LESSON NOTES

THIRD TERM

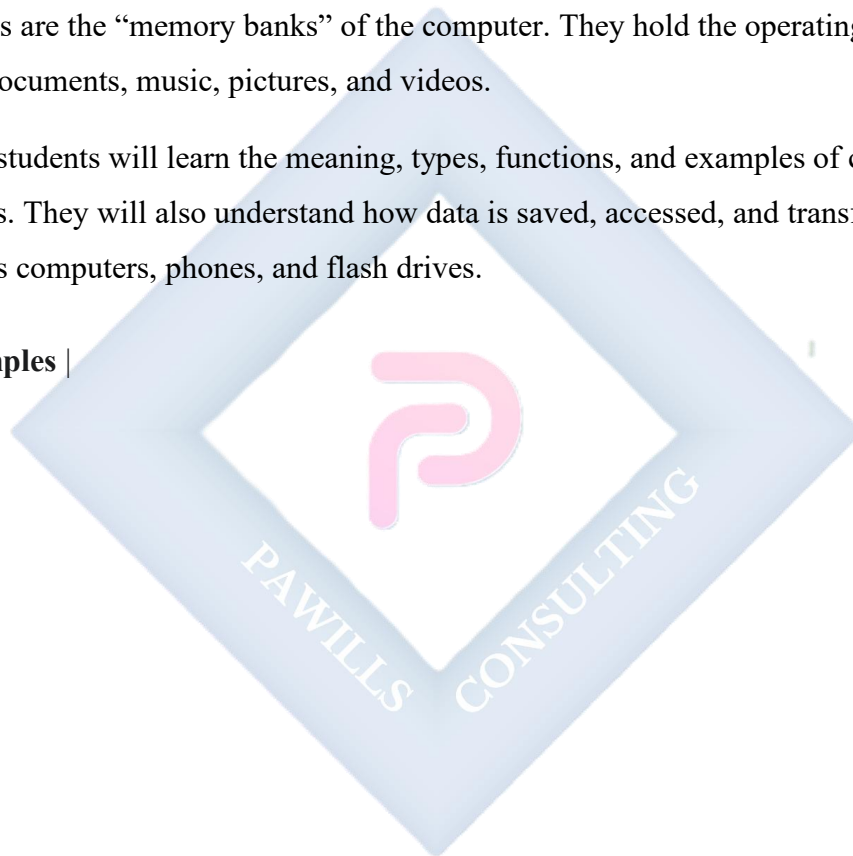
WEEK 1: COMPUTER STORAGE DEVICES

Introduction

Every computer needs a way to store data, instructions, and files so that information can be used now or later. This ability to save and retrieve data is made possible through **storage devices**. Storage devices are the “memory banks” of the computer. They hold the operating system, applications, documents, music, pictures, and videos.

In this lesson, students will learn the meaning, types, functions, and examples of computer storage devices. They will also understand how data is saved, accessed, and transferred between devices such as computers, phones, and flash drives.

Storage Examples |



COMPUTER STORAGE DEVICES: EXAMPLES

1. INTERNAL STORAGE



HDD (HARD DISK DRIVE)
High capacity, slower, mechanical



SSD SOLID STATE
Fast, durable, no moving parts

2. EXTERNAL STORAGE



USB FLASH DRIVE
Portable, convenient



MICRO SD CARD
Compact, mobile devices



STORING DATA: INTERNAL VS. EXTERNAL

1. Internal: For system & large files.
2. External: For backup & transfer.
3. All store digital information.



Meaning of Storage Devices

A **storage device** is a piece of computer hardware used to record and hold data permanently or temporarily. It stores digital information such as text, pictures, audio, videos, and programs.

In simple terms, it is the **part of the computer where data is saved** so that it can be used again when needed. Without storage devices, the computer would forget all data once it is turned off.

Types of Storage Devices

Computer storage is divided into three main types based on how they are used and how long they can hold data:

1. Primary Storage (Main Memory)

This is the internal memory of the computer that holds data temporarily while the computer is working.

It is directly accessible by the Central Processing Unit (CPU).

Examples:

- **RAM (Random Access Memory):** Temporary memory that stores information being used by the computer. All data is lost when power is turned off.
- **ROM (Read Only Memory):** Permanent memory that holds important startup instructions needed when the computer boots. Data here cannot be changed easily.

Characteristics of Primary Storage:

- Fast in operation.
- Small in capacity.
- Volatile (RAM loses data when power is off).
- Found inside the system unit.

2. Secondary Storage (External or Auxiliary Storage)

This storage holds data permanently. It is used when the primary memory is not enough. It stores operating systems, software, and user files.

Examples:

- **Hard Disk Drive (HDD):** A magnetic disk that stores large amounts of data. It is usually fixed inside the computer.

- **Solid State Drive (SSD):** A newer and faster storage device without moving parts. It loads programs more quickly than HDD.

Characteristics of Secondary Storage:

- Non-volatile (data remains after power is off).
- Large storage capacity.
- Slower than RAM but faster than optical media.
- Used for long-term storage.

3. Tertiary or Removable Storage

These are portable devices that can be connected or removed from the computer easily. They are useful for **backups, data transfer, and portability**.

Examples:

- **Flash Drives (USB Drives):** Small, portable devices used for transferring data.
- **Memory Cards (SD Cards):** Used in mobile phones, digital cameras, and tablets.
- **Optical Discs (CDs, DVDs, Blu-ray):** Used to store music, movies, or software.
- **External Hard Drives:** Larger portable drives for storing backups and large files.

Functions of Storage Devices

1. To save data, programs, and documents for future use.
2. To retain information even when the computer is switched off.
3. To allow users to transfer files from one device to another.
4. To store backups in case data is lost.
5. To support multimedia and entertainment by storing pictures, songs, and videos.

6. To enable fast retrieval of data by the processor.

Comparison Between Common Storage Devices

Device	Portability	Storage Capacity	Speed	Common Use
HDD	Fixed or external	High (500GB–2TB)	Moderate	File storage, operating systems
SSD	Fixed	High (256GB–1TB)	Very fast	Modern computers, gaming systems
Flash Drive	Portable	Medium (8GB–256GB)	Fast	File transfer and backup
Memory Card	Portable	Medium (2GB–128GB)	Fast	Phones, cameras
CD/DVD	Portable	Low (700MB–4.7GB)	Slow	Media storage

Importance of Storage Devices

1. They enable users to save and retrieve work easily.
2. They provide a permanent record of information.
3. They make data sharing and transfer possible between computers.
4. They protect important files through backup copies.
5. They support the computer's overall performance and functionality.
6. They allow multimedia use — saving music, movies, and photos.

Maintenance and Care of Storage Devices

1. Avoid exposure to dust, heat, or moisture.
2. Do not bend or scratch optical discs.
3. Safely eject USB drives before removing them.
4. Avoid dropping portable drives.
5. Keep devices away from magnets, which can erase data.
6. Label storage devices properly for easy identification.
7. Use antivirus programs to prevent virus infection of storage media.

Summary

Storage devices are essential components of every computer. They store data, instructions, and applications for future use. There are three main types: primary, secondary, and tertiary storage. Examples include hard drives, SSDs, flash drives, and memory cards. Proper handling and maintenance ensure long-lasting performance and data safety.

Evaluation

1. Define a computer storage device.
2. Mention three types of storage devices.
3. Give two examples each of primary, secondary, and tertiary storage.
4. State three functions of storage devices.
5. Explain the difference between volatile and non-volatile memory.

Assignment

1. List and describe five storage devices used in computers and phones.

2. Differentiate between HDD and SSD in terms of structure and speed.
3. Write four ways of caring for or maintaining storage devices.



WEEK 2: GSM NETWORKS

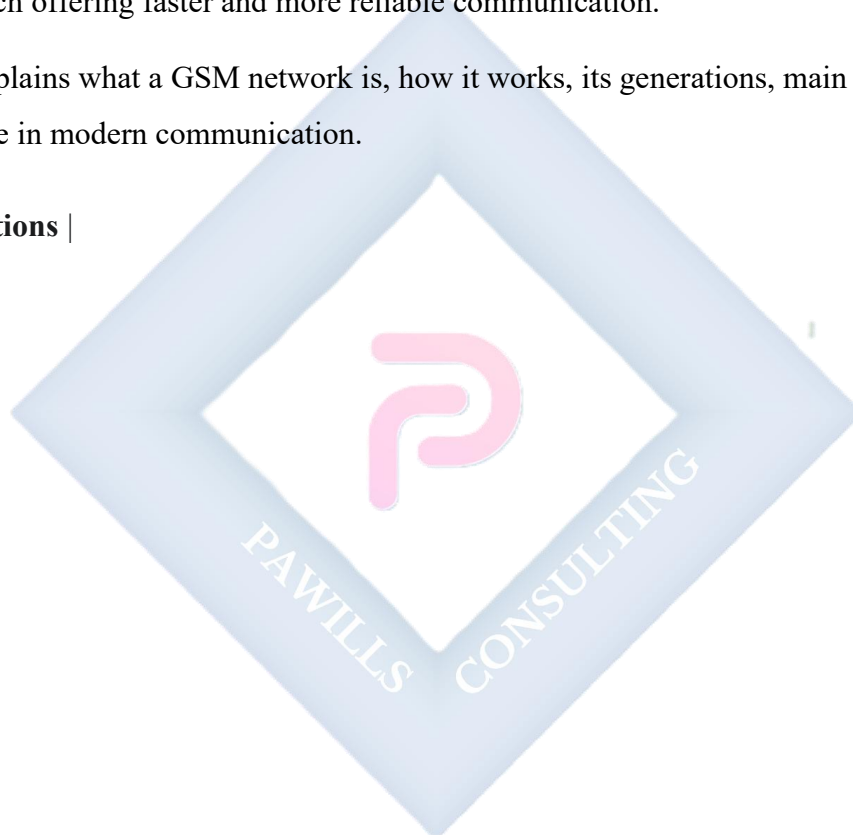
Introduction

The Global System for Mobile Communication (GSM) is the technology that allows mobile phones to send and receive calls, text messages, and internet data. It is the standard network system used by most mobile operators around the world.

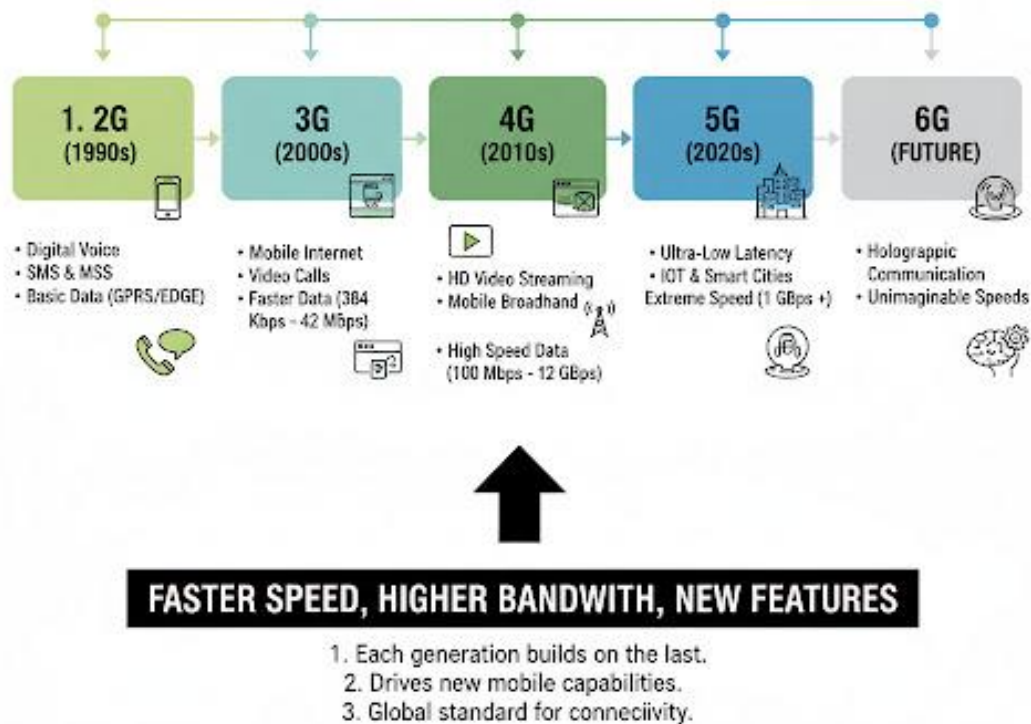
GSM networks make it possible for people to connect wirelessly, without physical cables. Over the years, mobile networks have developed through several generations — **2G, 3G, 4G, and now 5G** — each offering faster and more reliable communication.

This lesson explains what a GSM network is, how it works, its generations, main components, and importance in modern communication.

GSM Generations |



EVOLUTION OF GSM NETWORK GENERATIONS



Meaning of GSM Network

GSM stands for **Global System for Mobile Communication**. It is a digital network system that enables mobile devices (phones, tablets, modems, etc.) to communicate through radio waves.

In a GSM network, signals are transmitted between a user's mobile phone and nearby network towers (base stations). These towers connect to a central system that routes calls, messages, and data to their correct destinations.

Therefore, GSM is what allows your phone to make calls, send text messages, and access mobile internet services.

Basic Features of a GSM Network

1. Subscriber Identity Module (SIM Card):

- A small chip inserted into the phone.
- Stores user identity, phone number, and network data.

2. Cellular Coverage:

- The network is divided into small areas called *cells*, each with its own tower (Base Transceiver Station).

3. International Roaming:

- Allows users to connect to other GSM networks when traveling abroad.

4. Data and Voice Communication:

- Supports both phone calls and internet data services.

5. Security and Encryption:

- GSM uses encryption to protect user information and communication.

Generations of GSM Networks (2G – 5G)

GSM technology has evolved through different **generations**, each improving the speed, quality, and capabilities of mobile communication.

1. 2G (Second Generation)

- Introduced in the early 1990s.
- First to use digital signals instead of analog.
- Supported SMS (Short Message Service) and voice calls.

- Introduced slow internet using **GPRS (General Packet Radio Service)** and **EDGE (Enhanced Data Rates for GSM Evolution)**.
- Example networks: MTN 2G, Airtel 2G.

2. 3G (Third Generation)

- Introduced in the early 2000s.
- Improved internet speed and supported **video calling** and **multimedia messages (MMS)**.
- Enabled mobile web browsing, email, and online apps.
- Internet speed: up to **2 Mbps**.
- Example networks: Glo 3G, 9mobile 3G.

3. 4G (Fourth Generation)

- Launched around 2010.
- Provided **high-speed internet**, HD video streaming, online gaming, and fast downloads.
- Uses **LTE (Long Term Evolution)** technology.
- Internet speed: up to **100 Mbps** or more.
- Example: MTN 4G, Airtel 4G.

4. 5G (Fifth Generation)

- Latest and fastest mobile network technology.
- Supports **ultra-high-speed connectivity** and **low delay (latency)**.
- Used for smart devices, online learning, telemedicine, and Internet of Things (IoT).
- Internet speed: up to **10 Gbps** (100 times faster than 4G).
- Example: MTN 5G, Airtel 5G.

Components of a GSM Network

A GSM network is made up of several important parts that work together to send and receive information.

1. Mobile Station (MS):

- Consists of the **mobile phone** and **SIM card** used by the subscriber.
- Sends and receives signals to and from the network.

2. Base Station Subsystem (BSS):

- Contains **Base Transceiver Station (BTS)** and **Base Station Controller (BSC)**.
- Handles radio communication with phones and controls several BTS units.

3. Network Switching Subsystem (NSS):

- The “brain” of the network that routes calls and manages connections.
- Includes the **Mobile Switching Centre (MSC)**, **Home Location Register (HLR)**, and **Visitor Location Register (VLR)**.

4. Operation Support Subsystem (OSS):

- Manages, monitors, and maintains the network.
- Ensures smooth operation and quick repair of faults.

Importance of GSM Networks

1. Enables global voice and video communication.
2. Provides mobile internet access for browsing, streaming, and online learning.
3. Supports mobile banking and e-commerce.
4. Facilitates emergency communication and quick access to information.
5. Connects rural and urban communities.
6. Encourages business growth and social development.

Differences Between the Generations of GSM Networks

Generation	Key Features	Speed	Services Provided
2G	Digital voice, SMS, low-speed data	Slow (64 kbps)	Calls, texts
3G	Mobile internet, video calls	Medium (2 Mbps)	Internet, video calls
4G	High-speed internet, HD streaming	Fast (100 Mbps)	Online gaming, streaming
5G	Ultra-fast, low latency, IoT support	Very Fast (10 Gbps)	Smart devices, automation

Advantages of GSM Networks

1. Provides reliable and wide communication coverage.
2. Allows international roaming and connectivity.
3. Ensures security of user data through encryption.
4. Reduces cost of communication through SMS and data.
5. Supports multiple users within the same network.

Common Problems in GSM Networks

1. **Weak Signal:** Caused by distance from the tower or physical obstructions.

2. **Call Drops:** Occur due to network congestion or switching errors.
3. **Slow Internet:** Caused by poor coverage or too many users on one network.
4. **SIM Registration Error:** Due to improper activation or expired SIM.

Safety and Responsible Use of GSM Networks

1. Avoid using phones while driving.
2. Keep mobile phones away from water and extreme heat.
3. Do not share personal information over unknown calls or messages.
4. Avoid spending too much time online to prevent addiction.
5. Report lost or stolen SIM cards to the network provider immediately.

Summary

GSM networks are systems that connect mobile phones for voice and data communication through radio signals. They have evolved from 2G to 5G, with each generation offering faster speed, better quality, and more advanced features. The main components of a GSM network include the mobile station, base station subsystem, network switching subsystem, and operation support subsystem. GSM remains the backbone of global mobile communication today.

Evaluation

1. What does GSM stand for?
2. Mention and explain four generations of GSM networks.
3. State three components of a GSM network.
4. List two uses of GSM networks.
5. Explain one major difference between 4G and 5G.

Assignment

1. Draw and label a simple diagram showing how GSM connects a mobile phone to the network tower.
2. List four advantages of GSM networks to individuals and society.
3. Describe two common GSM network problems and how they can be solved.



WEEK 3: ASSEMBLING A SIMPLE COMPUTER

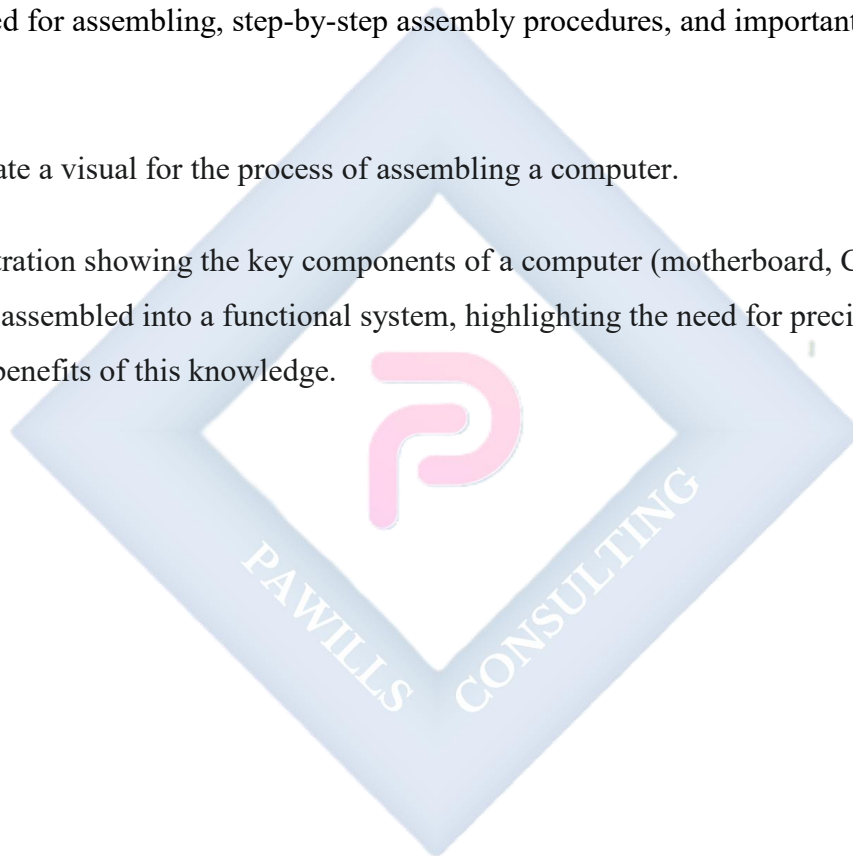
Introduction

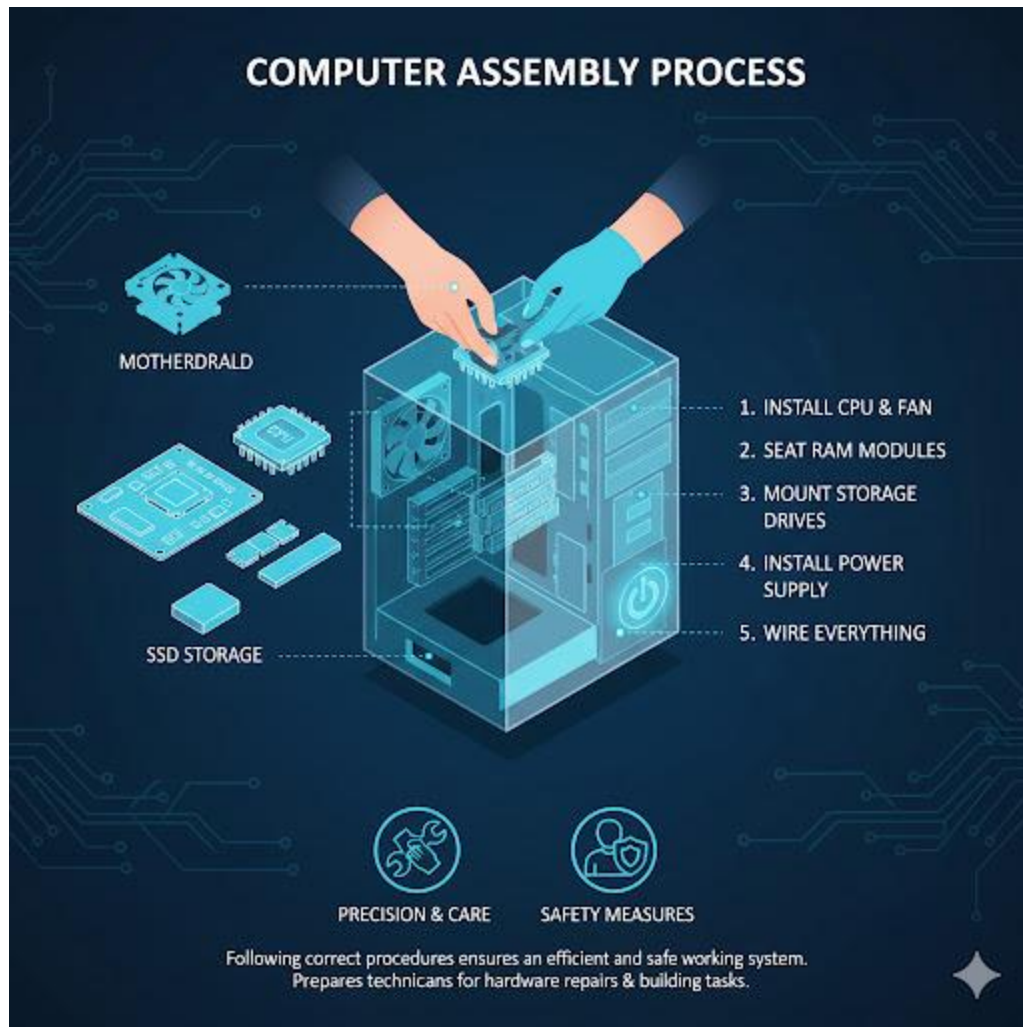
Every computer is made up of several hardware components that must be correctly connected before it can function. The process of putting these different parts together to form a working computer system is known as **computer assembling**.

Assembling a computer requires careful handling, the right tools, and understanding of how each part fits into the system. This lesson will teach students how to identify computer components, the tools needed for assembling, step-by-step assembly procedures, and important safety precautions.

Okay, let's create a visual for the process of assembling a computer.

Here's an illustration showing the key components of a computer (motherboard, CPU, RAM, storage) being assembled into a functional system, highlighting the need for precision and proper tools, and the benefits of this knowledge.





Meaning of Computer Assembling

Computer assembling is the process of **joining various hardware parts** of a computer together to form a complete and functional system. It involves installing components such as the motherboard, processor, memory (RAM), hard drive, and power supply into the computer case, and then connecting peripheral devices like the monitor, keyboard, and mouse.

A well-assembled computer ensures efficiency, durability, and good performance.

Major Components Needed for Computer Assembly

1. **System Unit (Computer Case):**

- The enclosure that houses all the internal components of the computer.
- Protects parts from dust, heat, and damage.

2. Motherboard:

- The main circuit board that connects all components together.
- It holds the CPU, RAM, storage devices, and expansion cards.
- Acts as the “heart” of the computer.

3. Processor (CPU – Central Processing Unit):

- The brain of the computer that performs calculations and executes commands.
- Usually installed on the motherboard with a cooling fan on top.

4. Memory (RAM – Random Access Memory):

- A temporary storage device used by the CPU to hold data currently being processed.
- The more RAM a computer has, the faster it performs.

5. Storage Device (Hard Disk Drive or Solid State Drive):

- Stores the operating system, programs, and user files.
- HDDs have moving parts, while SSDs are faster and use flash memory.

6. Power Supply Unit (PSU):

- Converts electricity from AC (from wall socket) to DC (used by the computer).
- Supplies power to all components in the system unit.

7. Cooling Fan or Heat Sink:

- Prevents the processor and other components from overheating.

8. Optical Drive (optional):

- Used to read or write data on CDs or DVDs.

9. Peripheral Devices:

- Devices connected externally such as the monitor, keyboard, mouse, and printer.

Tools Required for Computer Assembling

1. **Screwdrivers (Phillips and Flat Head):** For tightening or loosening screws.
2. **Anti-static Wrist Strap:** Prevents static electricity from damaging sensitive parts.
3. **Tweezers:** Helps handle small screws and components.
4. **Cable Ties:** Keeps wires organized inside the system unit.
5. **Thermal Paste:** Applied between the CPU and heat sink for better cooling.
6. **Brush and Blower:** Used to clean dust from computer parts.
7. **Torchlight or Lamp:** Provides clear lighting during installation.

Step-by-Step Procedure for Assembling a Simple Computer

Step 1: Prepare the Work Area

- Ensure the workspace is clean, dry, and well-lit.
- Place all components and tools neatly on a non-metallic surface.
- Wear an anti-static wrist strap to prevent electrostatic discharge.

Step 2: Open the System Unit Case

- Use a screwdriver to remove the side panels of the CPU case.
- Keep the screws in a safe place.

Step 3: Install the Power Supply Unit (PSU)

- Fix the power supply at the back of the case using screws.
- Ensure the fan faces outward for proper ventilation.

Step 4: Mount the Motherboard

- Align the motherboard with the standoffs (metal spacers) in the case.
- Screw it firmly in place but do not overtighten.

Step 5: Install the Processor (CPU)

- Gently open the CPU socket lever on the motherboard.
- Place the CPU correctly (match the arrow indicator).
- Close the lever to secure it.
- Apply a small amount of thermal paste on top of the CPU.
- Mount the cooling fan or heat sink above it.

Step 6: Insert the Memory (RAM)

- Locate the RAM slots on the motherboard.
- Insert the RAM module carefully until it clicks into place.

Step 7: Fix the Storage Drive (HDD/SSD)

- Slide the drive into its bay and screw it firmly.
- Connect power and data cables (SATA) from the motherboard and PSU.

Step 8: Connect Front Panel and Power Cables

- Attach the small front panel connectors (power switch, reset, LED).
- Connect the PSU's main power cable to the motherboard (24-pin connector).
- Attach the CPU power connector (4 or 8-pin).

Step 9: Install Other Components (if any)

- Add expansion cards such as graphics card, sound card, or network card.

Step 10: Close the Case

- Organize cables using cable ties.

- Replace the side panels and tighten screws.

Step 11: Connect External Devices

- Plug in the monitor, keyboard, mouse, and power cable.
- Connect the system to a power source.

Step 12: Power On and Test

- Press the power button.
- Check that the fan spins and the computer displays information on the screen.
- If successful, install the operating system and drivers.

Safety Precautions During Computer Assembly

1. Always switch off and unplug the power before starting.
2. Avoid touching sensitive parts with wet or oily hands.
3. Handle all components gently by the edges.
4. Use an anti-static wrist strap to prevent damage from static electricity.
5. Keep screws and small parts in labeled containers.
6. Do not force components into slots.
7. Ensure all cables are correctly connected before turning on the computer.
8. Work in a clean, dry, and static-free environment.

Common Mistakes to Avoid

1. Using too much thermal paste on the CPU.
2. Forgetting to connect the CPU or motherboard power cable.
3. Mixing up screws or dropping them inside the case.

4. Mounting the motherboard without standoffs, causing short circuit.
5. Forgetting to connect front panel connectors (power/reset switch).

Benefits of Learning Computer Assembly

1. Builds practical computer repair and maintenance skills.
2. Helps understand how each component works.
3. Reduces the cost of buying pre-assembled computers.
4. Encourages problem-solving and technical confidence.
5. Enables students to upgrade or repair systems themselves.

Summary

Assembling a computer means joining all necessary hardware components — such as the motherboard, CPU, RAM, and storage — to form a complete working system. The process requires precision, care, and proper tools. Following correct procedures and safety measures ensures the computer works efficiently and safely.

Proper assembly knowledge prepares students to handle hardware repairs, maintenance, and computer building tasks confidently.

Evaluation

1. Define computer assembling.
2. List five internal components required to assemble a computer.
3. Mention three tools used in assembling a computer.
4. Explain why thermal paste is applied during computer assembly.
5. State three safety precautions to observe when assembling a computer.

Assignment

1. Write out in order the **twelve steps** involved in assembling a simple computer.
2. List **five common mistakes** to avoid during computer assembly.
3. Explain **three reasons** why a student should learn how to assemble a computer.



WEEK 4: COMMON GSM PROBLEMS

Introduction

The Global System for Mobile Communication (GSM) has become a vital part of modern life. It enables people to make calls, send messages, browse the internet, and perform mobile banking. However, just like other electronic devices, GSM phones can develop faults or experience problems that affect their performance.

In this lesson, students will learn the **common GSM problems**, their **causes**, and **possible solutions**. They will also understand how to properly care for and maintain mobile phones to ensure long-lasting use.



Meaning of GSM Problems

GSM problems refer to faults, malfunctions, or irregularities that prevent a mobile phone from functioning correctly. These problems can occur in the **hardware** (physical parts) or **software** (programs and system files) of the phone.

Understanding GSM problems helps users know how to troubleshoot minor faults and when to seek professional repair.

Types of GSM Problems

GSM problems are generally classified into **hardware problems** and **software problems**.



1. Hardware Problems

Hardware problems are faults that occur in the **physical components** of a phone. They usually result from careless handling, falls, exposure to water, or electrical damage.

Examples include:

- **Broken screen:** Usually caused by dropping the phone.

- **Battery not charging:** May result from damaged charging port, faulty charger, or bad battery.
- **Speaker or earpiece not working:** Could be due to dust, moisture, or loose connection.
- **Keypad or touch not responding:** Occurs when the screen or keypad gets damaged.
- **Microphone not functioning:** The person at the other end cannot hear the user clearly.
- **No network signal:** May be due to faulty antenna or damaged SIM tray.

2. Software Problems

Software problems are caused by faults in the phone's **operating system, applications, or memory**.

Examples include:

- **Phone freezing or hanging:** Caused by too many applications running or insufficient memory.
- **Slow performance:** Due to excessive files, viruses, or outdated software.
- **Automatic restart:** Often results from system corruption or malware attack.
- **Application not opening or crashing:** Caused by faulty app installation.
- **Security lock forgotten:** The user cannot access the phone because of password or pattern lock.

Common Causes of GSM Problems

COMMON CAUSES OF GSM PROBLEMS



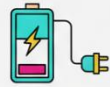
PHYSICAL DAMAGE

Drops, water, and heat.



POWER SURGE

Sudden voltage changes.



OVERCHARGING

Reduces battery lifespan.



VIRUS OR MALWARE

Unsafe downloads/websites



OVERLOADING MEMORY

Too many apps, photos, files.



FAKE ACCESSORIES

Damages charging system

1. **Physical Damage:** Dropping the phone or exposing it to water or heat.
2. **Overcharging:** Reduces battery lifespan.
3. **Use of Fake Chargers or Accessories:** Can damage the charging system or battery.
4. **Virus or Malware Attack:** From downloading unsafe files or visiting untrusted websites.
5. **Lack of Regular Software Updates:** Causes poor performance.
6. **Overloading Memory:** Storing too many apps, videos, or pictures.
7. **Power Surge:** Sudden voltage changes can damage internal circuits.
8. **Improper Handling:** Rough use or pressure on the screen.
9. **Factory Defects:** Some problems come from the manufacturer.
10. **Exposure to Dust and Moisture:** Reduces the lifespan of phone components.

Troubleshooting Common GSM Problems

Troubleshooting means identifying and correcting faults in a system. Below are common GSM problems and how to fix them:

Problem	Possible Cause	Solution
Phone not charging	Faulty charger, bad charging port, or damaged battery	Try another charger; clean or replace charging port; change battery if needed
Screen not displaying	Broken or disconnected screen cable	Replace the screen or reconnect the cable
Phone not coming on	Dead battery or motherboard issue	Charge battery or visit a technician
No sound during call	Faulty earpiece or speaker	Replace earpiece or speaker
No network signal	SIM not inserted properly or antenna problem	Reinsert SIM card; check network settings
Phone hangs or freezes	Full memory or too many apps	Delete unused apps and clear cache
Apps closing automatically	Corrupt software	Reinstall app or reset phone
Phone overheating	Long usage or overcharging	Allow phone to cool; avoid heavy apps
Battery draining quickly	Background apps or old battery	Close background apps; replace battery
Phone infected by virus	Unsafe downloads	Install antivirus or reset phone

TROUBLESHOOING COMMON GSM PROBLEMS

PROBLEM & CAUSE	SOLUTION
 SCREEN NOT DISPLAYING Cause: Broken cable	 Replace the screen or reonect cable
 PHONE NOT CHARGING Cause: Faulty charger/port	 Try another charger; Clean port
 NO NETWORK SIGNAL Cause: SIM not inserted	 Reinesrt SIM card; Check network settings

Preventive Measures and Care for GSM Phones

1. **Handle the phone carefully** to avoid drops and shocks.
2. **Keep the phone away from water, moisture, and heat.**
3. **Avoid overcharging;** unplug the charger once full.
4. **Use only original chargers and batteries.**
5. **Clean the phone regularly** using a dry soft cloth.
6. **Install antivirus software** to protect against malware.
7. **Update software regularly** to improve performance.
8. **Avoid downloading from untrusted websites.**
9. **Use a screen guard and protective case** to prevent scratches.

10. Store the phone in a safe and dry place when not in use.

Importance of Knowing GSM Problems

1. Helps users detect faults early and prevent bigger damage.
2. Saves money by fixing simple issues without a technician.
3. Promotes proper handling and maintenance culture.
4. Builds technical and problem-solving skills in students.
5. Encourages responsible use of mobile technology.



Summary

GSM problems can be **hardware** or **software** related. Common issues include battery not charging, screen damage, hanging, poor network, or virus infection.

Causes range from careless handling and overcharging to malware attacks.

Simple troubleshooting like restarting, cleaning, or resetting the phone can fix minor problems,

while serious ones should be handled by a technician.

Proper care, regular updates, and use of original accessories help prevent many GSM faults.

Evaluation

1. Define GSM problems.
2. List four examples of hardware problems in mobile phones.
3. Mention three causes of GSM software problems.
4. Explain two preventive measures for GSM phone faults.
5. What does troubleshooting mean in GSM maintenance?

Assignment

1. List **five common GSM problems** and their **possible causes**.
2. Explain **three ways** of preventing damage to your mobile phone.
3. Describe how to troubleshoot a phone that **refuses to power on**.

WEEK 6: PRACTICAL – REPLACING GSM PARTS (BATTERY, SCREEN, KEYPAD)

Introduction

Every GSM phone is made up of several hardware components that can wear out, get damaged, or stop working properly. As a GSM technician or student learning phone repair, it is important to know how to replace faulty parts correctly and safely.

This practical lesson focuses on **replacing three common GSM components — battery, screen, and keypad** — which are among the most frequently damaged or replaced parts in mobile phones.

Students will learn how to identify these parts, use the right tools, follow proper procedures, and observe safety measures when replacing them.

Meaning of GSM Parts Replacement

GSM parts replacement means removing damaged or faulty phone components and fixing new or functional ones in their place.

It is a vital part of phone repair that restores the device to proper working condition.

This process requires careful handling, correct tools, and knowledge of how each part connects to the phone's motherboard and casing.

Reasons for Replacing GSM Parts

1. Damage due to accidental falls or impact.
2. Wear and tear from long usage.
3. Water or moisture damage.
4. Electrical faults or overcharging.
5. Replacement of swollen or dead battery.
6. To improve phone performance (screen brightness, touch sensitivity, etc.).
7. To restore physical appearance after damage.

8. Poor response or malfunctioning keypad.
9. Defective touch or broken display.

Tools Required for GSM Parts Replacement



1. **Small screwdrivers (Phillips and flat head)** – for removing screws.
2. **Plastic pry tools or openers** – for separating phone casing safely.
3. **Tweezers** – for handling tiny components.
4. **Suction cup** – for removing screen panels.

5. **Adhesive glue or double-sided tape** – for attaching new screens.
6. **Soldering iron and solder wire** – for fixing or removing electronic parts.
7. **Anti-static wrist strap** – to prevent electrostatic discharge.
8. **Magnifying glass or lamp** – for better visibility of small parts.
9. **Brush and blower** – for cleaning the phone before and after replacement.
10. **Replacement parts** – battery, screen, keypad.

Main Components to Be Replaced

1. Battery

The battery supplies power to the phone. It may be **removable** or **non-removable (inbuilt)**.

Signs of a faulty battery:

- Drains quickly
- Swollen or overheats
- Does not charge or power the phone
- Causes phone to restart frequently

2. Screen

The screen displays images and allows touch input. It consists of the **LCD (display)** and **touchscreen (digitizer)**.

Signs of a faulty screen:

- Cracks or lines on the display
- Touch not responding
- Blank or flickering display

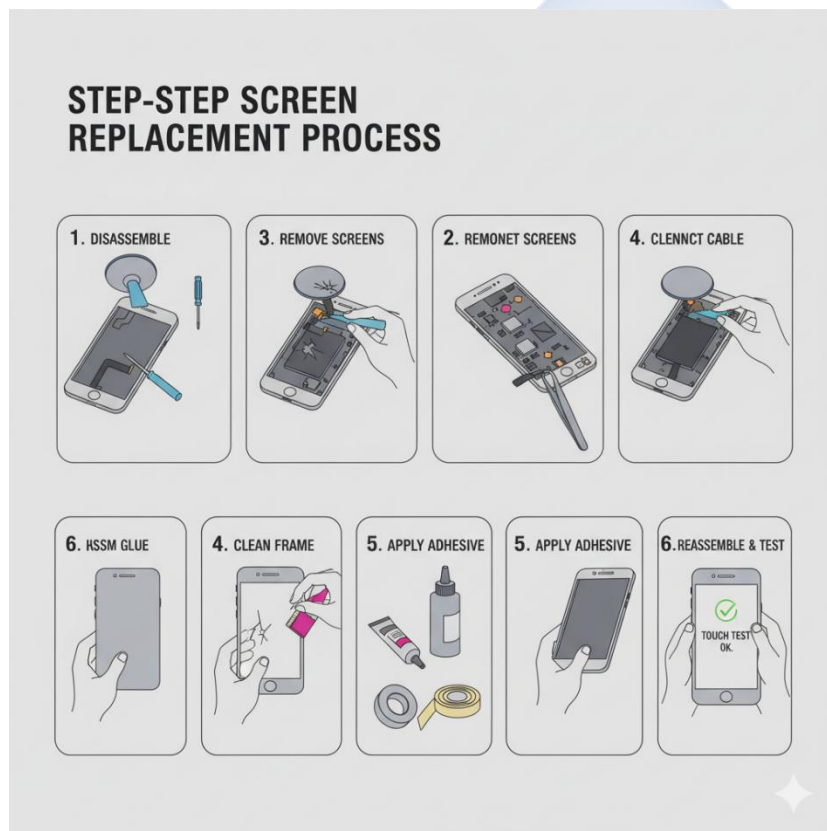
3. Keypad

Used in button phones for typing numbers and letters.

Signs of a faulty keypad:

- Some buttons not responding
- Keys stuck or hard to press
- Dirt or moisture under the buttons

Step-by-Step Procedures for Replacing GSM Parts



A. Replacing the Battery

1. **Turn off the phone completely.**
2. **Remove the back cover** using a pry tool or fingernail.

3. **Take out the old battery** carefully.
4. **Inspect the battery compartment** for dust or corrosion.
5. **Insert the new battery** ensuring the terminals align correctly.
6. **Replace the back cover** securely.
7. **Switch on the phone** to test if it powers and charges correctly.

For non-removable batteries:

- Unscrew the casing and disconnect the battery connector from the motherboard.
- Replace it with a new one and reconnect properly before reassembling.

B. Replacing the Screen

1. **Switch off the phone and remove SIM and memory cards.**
2. **Use a screwdriver** to remove all screws securing the back cover.
3. **Separate the casing** using a plastic pry tool.
4. **Disconnect the screen connector** from the motherboard using tweezers.
5. **Gently remove the broken screen** using a suction cup.
6. **Clean the surface** and remove old adhesive.
7. **Attach the new screen** with glue or tape.
8. **Reconnect the screen cable** and reassemble the phone.
9. **Switch on the phone** and test the display and touch functions.

C. Replacing the Keypad

1. **Switch off the phone and remove the battery.**
2. **Unscrew the casing** and separate the front housing.

3. **Remove the old keypad** carefully.
4. **Clean the keypad area** with a soft brush or cotton swab.
5. **Insert the new keypad** correctly into position.
6. **Reassemble the phone** and tighten all screws.
7. **Power on the phone** to test the buttons.

Safety Precautions During GSM Parts Replacement

1. Always **switch off the phone** before opening it.
2. Use **appropriate tools** for each task.
3. **Do not use excessive force** when removing parts.
4. **Wear an anti-static wrist strap** to avoid damaging components.
5. Keep **screws and small parts organized** in labeled containers.
6. Avoid **touching the circuit board** with wet or oily hands.
7. **Handle screens with care** to prevent cracks.
8. **Disconnect the battery** before soldering any part.
9. Work on a **clean and well-lit surface**.
10. Always **test the phone** before reassembling completely.

Common Mistakes to Avoid

1. Using the wrong screwdriver size.
2. Forgetting to disconnect the battery before repair.
3. Misplacing screws or mixing their positions.
4. Applying too much glue on screens.

5. Pulling connectors roughly and damaging them.
6. Failing to test parts before reassembly.
7. Working without grounding protection (static wrist strap).
8. Ignoring phone cleanliness during reassembly.

Benefits of Learning GSM Parts Replacement

- Builds confidence in handling phone hardware.
- Helps identify and repair common GSM faults.
- Saves cost by fixing simple faults independently.
- Encourages neatness, precision, and patience.
- Improves students' practical and technical competence.
- Provides foundational skills for future technicians.

Summary

Replacing GSM parts is a practical process that involves removing damaged components such as batteries, screens, or keypads and fixing new ones carefully.

It requires using the right tools, following safety measures, and working systematically to avoid damage.

Proper replacement improves the phone's performance and extends its lifespan.

Mastering this skill is essential for every student learning GSM repair.

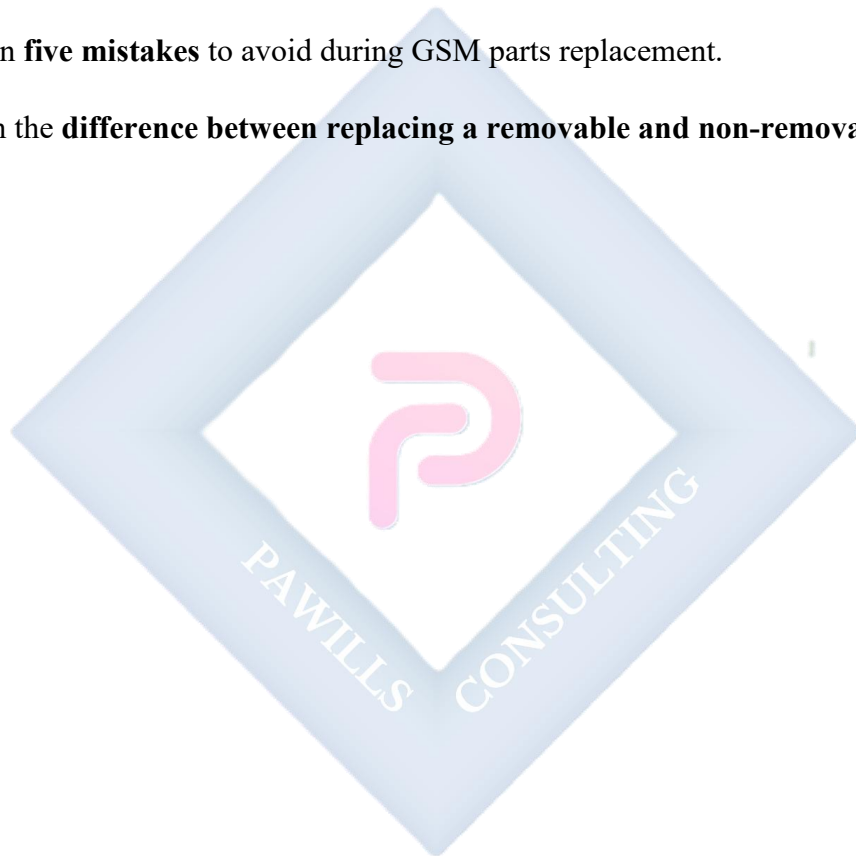
Evaluation

1. Define GSM parts replacement.
2. Mention three GSM components that are commonly replaced.

3. List five tools needed for replacing phone parts.
4. State four safety precautions to observe during GSM parts replacement.
5. Explain why it is important to test a phone after repair.

Assignment

1. List **seven steps** involved in replacing a phone screen.
2. Mention **five mistakes** to avoid during GSM parts replacement.
3. Explain the **difference between replacing a removable and non-removable battery**.



WEEK 8: INTRODUCTION TO SOLDERING

Introduction

In electronics and GSM repair, **soldering** is one of the most important practical skills every technician must master. It involves joining two or more electrical parts together using a melted metal called *solder*.

Soldering helps in connecting components firmly on a circuit board to ensure smooth flow of electric current.

In this lesson, students will learn the meaning of soldering, the tools and materials used, types of soldering, safety precautions, and basic techniques required to perform simple soldering tasks in GSM and computer hardware repairs.

Meaning of Soldering

Soldering is the process of joining two or more metal parts together by melting solder (a mixture of tin and lead or other alloys) around the connection point.

When the solder cools, it forms a strong electrical and mechanical joint that allows current to pass smoothly between the connected parts.

Soldering is used in **electronics, GSM repair, electrical wiring, and computer hardware** to fix or replace components on circuit boards.

Key Terms

- **Solder:** A fusible metal alloy (usually tin and lead) used to join metal surfaces.
- **Flux:** A chemical substance applied before soldering to clean metal surfaces and help the solder flow evenly.
- **Joint:** The point where two metals are connected by solder.
- **Desoldering:** The process of removing solder to detach a component.

Types of Soldering

There are **three main types of soldering**, depending on the temperature and tools used:

1. Soft Soldering

- Done at a low temperature (below 400°C).
- Commonly used in electronics and GSM repair.
- Uses a tin-lead solder alloy.
- Tools: Electric soldering iron, flux, and solder wire.

2. Hard Soldering

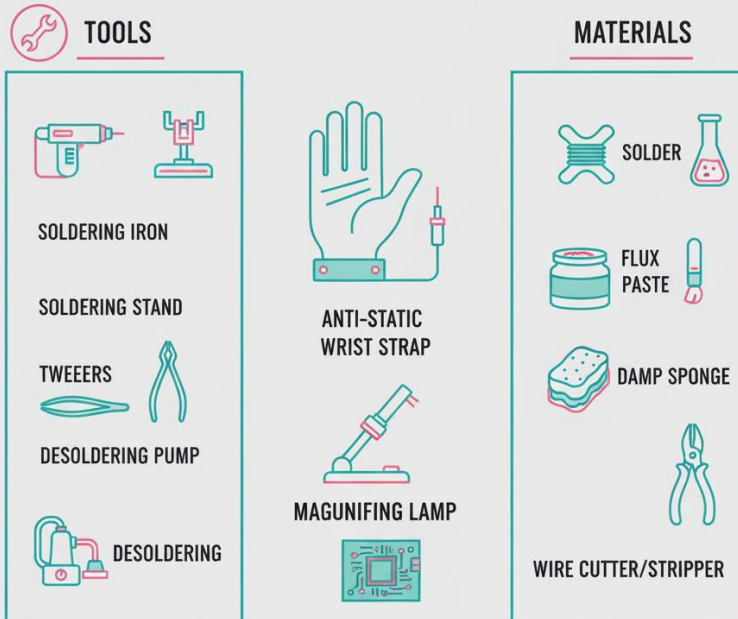
- Performed at a higher temperature using alloys such as brass or silver.
- Requires a blowtorch.
- Used for stronger joints in mechanical work.

3. Brazing

- Involves heating metals above 450°C using a filler metal.
- Common in industrial metal joining, not electronics.

Note: GSM and computer technicians mainly use **soft soldering**.

BASIC SOLDERING TOOLS AND MATERIALS



Tools and Materials Used in Soldering

Tool/Material	Function
Soldering Iron	Heats and melts solder.
Soldering Stand	Holds hot soldering iron safely when not in use.
Solder Wire	The metal alloy that melts to form the joint.
Flux Paste	Cleans surfaces and helps solder flow smoothly.
Sponge	Used to clean the soldering iron tip.
Tweezers	Hold small components during soldering.
Desoldering Pump	Removes excess solder from joints.

Wire Cutter/Stripper	Cuts and strips wires.
Magnifying Glass	Helps view small connections clearly.
Anti-static Wrist Strap	Prevents electric discharge that can damage components.

Steps in Soldering

1. Preparation

- Ensure your work area is clean and well-ventilated.
- Gather all soldering tools.
- Plug in and preheat the soldering iron.

2. Cleaning

- Clean the components and circuit board with flux or isopropyl alcohol.
- Wipe the soldering iron tip on a damp sponge to remove dirt.

3. Positioning

- Place the component in its proper hole or point on the circuit board.
- Hold it steady with tweezers or clips.

4. Applying Heat

- Touch the soldering iron tip to the joint area for a few seconds to heat it.

5. Applying Solder

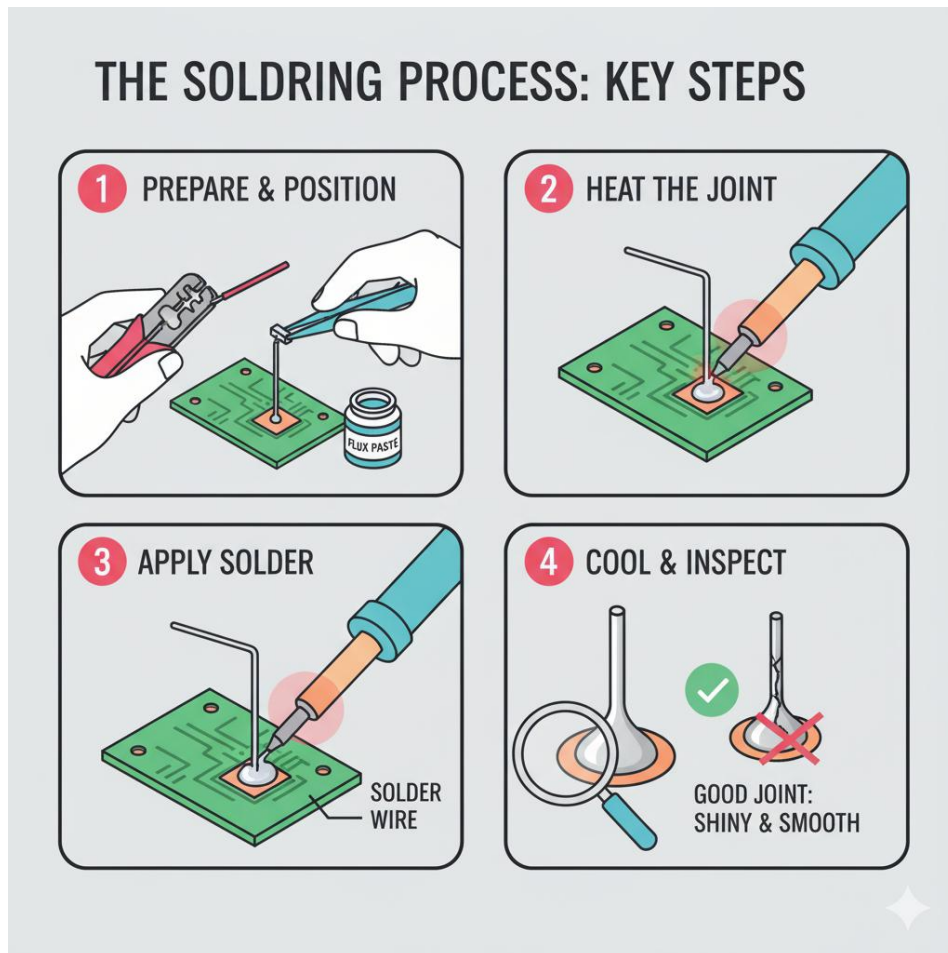
- Feed a small amount of solder wire to the joint (not directly to the iron tip).
- Allow the solder to melt and flow around the joint.

6. Cooling

- Remove the solder wire and iron.
- Allow the joint to cool naturally (do not blow air on it).

7. Inspection

- Check the joint to ensure it is shiny and smooth (a dull or cracked joint is bad).
- Trim off excess wire if necessary.



Safety Precautions During Soldering

1. Always work in a **well-ventilated area** to avoid inhaling fumes.
2. **Do not touch the metal part** of the soldering iron; it is very hot.
3. Place the soldering iron on its **stand when not in use**.
4. Keep **flammable materials** away from your workspace.

5. **Unplug the soldering iron** after use.
6. Use **protective goggles and gloves**.
7. Avoid eating or touching your face while soldering.
8. Handle the circuit board carefully to prevent static damage.
9. Dispose of solder waste safely.
10. Never apply too much solder — it can cause short circuits.

Common Soldering Mistakes

1. Using too much or too little solder.
2. Not cleaning the tip of the soldering iron.
3. Moving the joint before it cools.
4. Applying solder directly to the iron tip instead of the joint.
5. Overheating components and damaging them.
6. Working without ventilation or safety gear.
7. Using a faulty or low-quality soldering iron.

Importance of Soldering in GSM and Computer Repairs

1. Used to fix loose or broken wires on circuit boards.
2. Helps in replacing damaged components.
3. Ensures firm and durable electrical connections.
4. Used in assembling or modifying electronic devices.
5. Builds practical and professional repair skills.
6. Enhances precision and attention to detail.

Summary

Soldering is the process of joining metal parts by melting solder to form a permanent bond. It is widely used in GSM and computer repair work to attach or replace components on circuit boards.

Soft soldering is the type most suitable for electronics.

To achieve good results, one must use proper tools, follow correct procedures, and always observe safety rules.

A good solder joint is clean, shiny, and firmly holds the components together.

Evaluation

1. Define soldering.
2. Mention three types of soldering.
3. List five tools used in soldering.
4. State four safety precautions during soldering.
5. Explain why soldering is important in GSM repair.

Assignment

1. Describe in detail the **seven steps** involved in soldering a component.
2. List and explain **five common soldering mistakes** to avoid.
3. State **five differences** between soft and hard soldering.

WEEK 9: PRACTICAL – SOLDERING SIMPLE JOINTS

Introduction

In this lesson, students will apply what they learned about soldering by **performing basic soldering operations** on simple electronic joints.

Practical soldering is essential for anyone learning GSM and computer hardware repair because it helps in **attaching, replacing, and fixing components** on circuit boards such as resistors, capacitors, or wires.

This exercise teaches accuracy, patience, and careful handling of tools — all vital skills for a professional technician.

Meaning of a Soldered Joint

A **soldered joint** is the point where two or more electrical parts are connected together by melted solder.

When properly made, the joint allows smooth current flow and forms a **strong mechanical connection** between components.

Good solder joints appear shiny, clean, and smooth.

Bad solder joints look dull, rough, or cracked and can lead to **poor electrical contact or total failure** of a circuit.

Types of Solder Joints

There are several types of joints used in simple electronics work:

1. Lap Joint:

- Two wires overlap each other and are soldered together.
- Commonly used for connecting wires in GSM charging cables or earphones.

2. T-Joint:

- One wire is joined perpendicularly (in the form of a “T”) to another wire.

- Used when adding a branch connection.

3. **Western Union Joint (Twist Joint):**

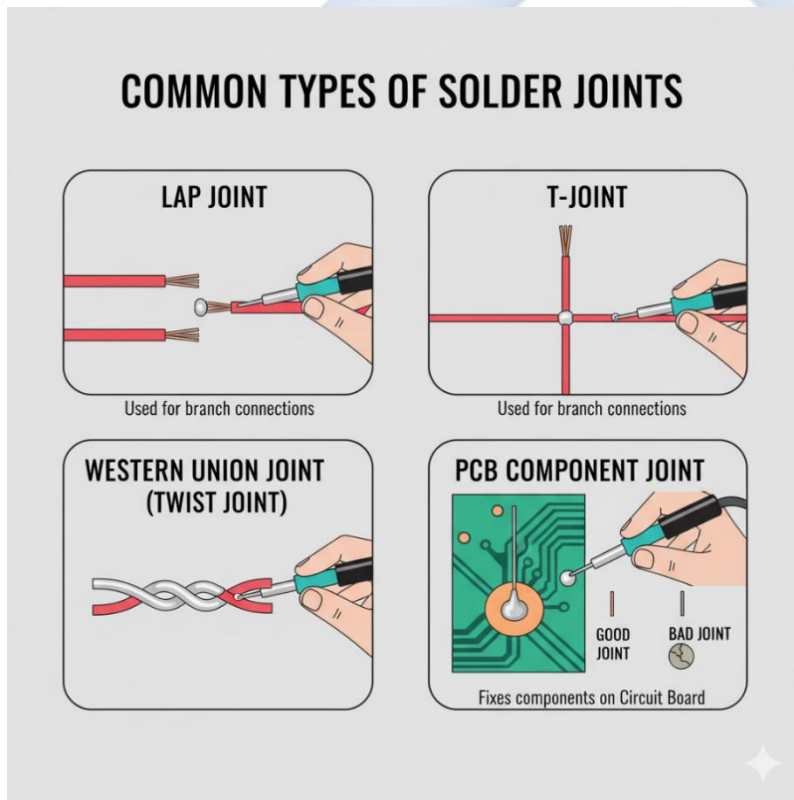
- Two wires are twisted together before soldering.
- Provides a strong mechanical bond.

4. **Butt Joint:**

- Two wires are joined end-to-end and soldered.
- Useful when extending broken or short wires.

5. **PCB Component Joint:**

- Used to fix or replace components (resistors, capacitors, ICs) on a **Printed Circuit Board (PCB)**.



Tools and Materials for the Practical

Tool/Material	Function
Soldering Iron	Heats and melts solder for joining metals.
Solder Wire	The metallic material that melts to form joints.
Soldering Stand	Holds the hot soldering iron safely.
Sponge	Cleans the soldering iron tip.
Flux	Cleans and prepares surfaces for soldering.
Wires	Used for making joints.
Circuit Board	Base for mounting components.
Tweezers	Hold small components in place.
Desoldering Pump	Removes unwanted solder.
Cutter/Stripper	Cuts and strips wire insulation.

Procedures for Soldering Simple Joints

A. Preparation

1. Clean and organize your work area.
2. Ensure proper ventilation to remove smoke or fumes.
3. Plug in and preheat the soldering iron.
4. Arrange all tools and materials neatly on the workbench.
5. Wear safety gear such as goggles and gloves.

B. Making a Simple Wire-to-Wire Joint

1. **Strip** about 1 cm of insulation from the ends of two wires.

2. **Twist** the exposed ends together (if using a twist joint).
3. **Apply flux** to the twisted ends to clean them.
4. **Heat the joint** gently with the soldering iron for a few seconds.
5. **Apply solder** to the joint until it flows smoothly over the wires.
6. **Remove the iron** and allow the joint to cool naturally.
7. **Inspect** the joint to ensure it is firm and shiny.

C. Mounting and Soldering Components on a Circuit Board

1. Identify the correct **holes** for the component on the PCB.
2. Insert the **component leads** through the holes.
3. Turn the board over and **bend the leads slightly** to hold it in place.
4. Heat the **joint area** with the soldering iron tip.
5. Feed solder into the joint — it should flow neatly around the lead and pad.
6. Remove the iron and let it **cool** for a few seconds.
7. Cut off **excess wire** from the component leads.
8. Inspect the joint — it should be smooth, shiny, and not touch any other track.

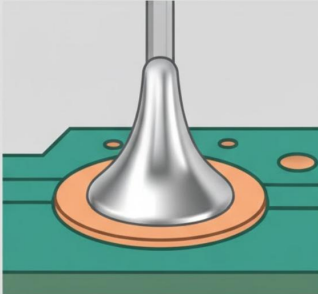
D. Testing the Joint

After completing soldering:

1. Use a **multimeter** to check for continuity.
2. Ensure there is no short circuit between tracks.
3. Confirm that components are firmly attached.
4. Power the device briefly to test operation (if safe to do so).

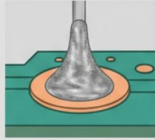
GOOD VS. BAD SOLDER JOINTS

GOOD SOLDER JOINT

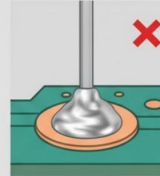


- **APPEARANCE:** Shiny, smooth, concave fillet.
- **ELECTRICAL:** Strong reliable connection

COMMON SOLDER FAULTS



✗ **COLD JOINT**



✗ **EXCESS SOLDER / BLOB JOINT**

CAUSE: Insufficient heat.
FIX: Reheat & add flux.
FIX: Use desoldering pump



✗ **SHORT CIRCUIT / SOLDER BRIDGE**

CAUSE: Too hot links two points. Solder links two points.
FIX: Remove with desolder with desoldering wick

Safety Measures During Practical

1. Always handle the soldering iron carefully; the tip is extremely hot.
2. Work in a **ventilated area** to avoid inhaling fumes.
3. Keep your **workbench clean and dry**.
4. Never touch the **metal part** of the soldering iron.
5. Hold small components with **tweezers**, not fingers.
6. Place the soldering iron on its **stand when not in use**.
7. Avoid dripping hot solder on your hands or clothes.
8. Unplug the soldering iron after use.
9. Keep flammable materials away.
10. Wash your hands after handling solder or flux.

Common Faults and How to Avoid Them

Fault	Cause	Prevention
Cold joint	Removing solder too quickly	Heat joint properly before removing iron
Dull or cracked joint	Dirty surfaces or poor solder	Clean parts and apply flux
Excess solder	Using too much solder wire	Apply small amount gradually
Short circuit	Solder bridges two joints	Use desoldering pump to remove excess
Loose component	Poor heating or movement before cooling	Hold part steady until solder cools

Importance of Practical Soldering Skills

1. Enables repair of damaged or loose connections.
2. Builds confidence in handling delicate components.
3. Encourages accuracy and neatness.
4. Helps students understand electronic assembly.
5. Prepares learners for real-life GSM and computer repair tasks.
6. Encourages safe and professional work habits.

Summary

Soldering simple joints is a vital practical skill in electronics and GSM repair.

The process involves preparing surfaces, applying heat, melting solder, and allowing the joint to cool.

Good solder joints are smooth, shiny, and firm, while bad ones are dull or weak.

Observing safety precautions and following correct procedures ensure high-quality and long-lasting results.

Evaluation

1. Define a soldered joint.
2. List three types of solder joints.
3. Mention four tools used in soldering.
4. State five safety rules to follow during soldering practical.
5. Explain two causes of poor solder joints.

Assignment

1. Describe **step-by-step** how to solder a component on a PCB.
2. Draw and label **two types of solder joints**.
3. List **five common soldering faults** and state how each can be corrected.

WEEK 10: PREVENTIVE MAINTENANCE

Introduction

Preventive maintenance refers to the **regular care and servicing** of computer systems and GSM phones to keep them working efficiently and to prevent unexpected breakdowns.

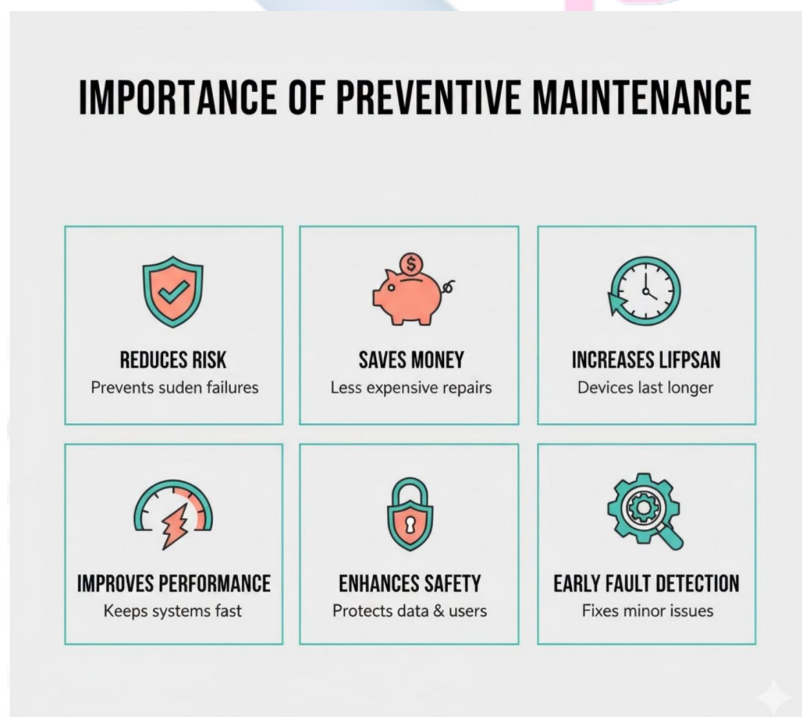
In this lesson, students will learn how to maintain both computer hardware and GSM devices through cleaning, inspection, testing, and proper handling.

Preventive maintenance is very important because it helps to extend the life of the equipment, reduce repair costs, and improve performance.

Meaning of Preventive Maintenance

Preventive maintenance is the **routine and planned servicing** of devices before a fault occurs. It involves periodic checking, cleaning, and replacing of weak parts to avoid total damage or sudden failure.

It is different from **corrective maintenance**, which is done **after** a fault has occurred.



Importance of Preventive Maintenance

1. Reduces the risk of sudden system or phone failure.
2. Saves money on expensive repairs and replacements.
3. Increases the lifespan of devices.
4. Keeps the system or phone running smoothly and faster.
5. Ensures user safety and data protection.
6. Helps in early detection of faults.
7. Reduces downtime in computer labs and repair shops.

Types of Preventive Maintenance

There are two main types:

1. **Hardware Preventive Maintenance:**
 - Deals with the **physical components** of the computer or phone.
 - Involves cleaning, inspecting, tightening connections, and replacing worn-out parts.
2. **Software Preventive Maintenance:**
 - Deals with maintaining the **programs and data** on a computer or smartphone.
 - Includes virus scanning, updating software, and deleting unnecessary files.

Preventive Maintenance in Computers

A. Hardware Maintenance Procedures

1. **Cleaning the System Unit:**
 - Use a soft brush or blower to remove dust.
 - Avoid using water or wet cloths on internal parts.

2. Keyboard and Mouse Cleaning:

- Turn them upside down and shake gently to remove dirt.
- Use a dry cloth or cotton swab for the keys.

3. Monitor and Screen Cleaning:

- Use a soft microfiber cloth to wipe gently.
- Do not use harsh chemicals or wet rags.

4. Power Supply Check:

- Ensure cables are properly connected and not frayed.
- Avoid overloading sockets.

5. Internal Component Check:

- Open the system unit occasionally to remove dust.
- Check RAM, fan, and cables for proper seating.



B. Software Maintenance Procedures

1. Install and update antivirus software regularly.
2. Delete temporary and unused files.
3. Perform disk cleanup and defragmentation.
4. Backup important data frequently.
5. Update the operating system and applications.
6. Avoid installing unverified programs.

Preventive Maintenance in GSM Devices

1. Regular Cleaning:

- Clean the phone screen and body with a soft dry cloth.
- Remove dirt and dust from charging and earphone ports.

2. Battery Care:

- Avoid overcharging or complete discharge.
- Use only the recommended charger.
- Keep the battery away from heat or moisture.

3. Screen Protection:

- Use a screen guard or pouch to prevent scratches and cracks.

4. Storage Management:

- Delete unwanted files and clear cache.
- Avoid installing too many apps.

5. Software Update:

- Keep the phone software up to date for better security and performance.

6. SIM and Memory Card Care:

- Handle carefully; avoid bending or scratching.
- Backup contacts and important data.

7. Environmental Precautions:

- Keep away from water, excessive heat, and magnetic fields.
- Avoid dropping or applying force to the phone.

Tools and Materials for Maintenance

1. Small screwdrivers
2. Cleaning brush and blower
3. Screen wipes and microfiber cloth
4. Cotton buds and isopropyl alcohol
5. Multimeter (for testing power)
6. USB cable and computer system (for updates and backup)
7. Software tools (antivirus, cleanup tools)

Steps in Preventive Maintenance (Computer or Phone)

1. **Inspect:** Check for physical damage, dust, or loose connections.
2. **Clean:** Use proper tools to remove dust and dirt.
3. **Test:** Power on and test basic functions.
4. **Update:** Run software or firmware updates.
5. **Backup:** Save important files to another storage device.
6. **Document:** Record maintenance date and activities for reference.

Safety Precautions During Maintenance

1. Always turn off and unplug devices before cleaning.
2. Use dry materials — avoid water or wet hands.
3. Handle internal components gently.
4. Avoid touching circuit boards with bare hands.
5. Use antistatic tools or wrist straps.
6. Store devices in safe, dry places.
7. Follow manufacturer's instructions.
8. Do not use metal tools when device is powered on.

Signs That Preventive Maintenance Is Needed

1. System or phone runs slower than usual.
2. Frequent freezing, crashing, or restarting.
3. Unusual noises or overheating.
4. Weak battery or poor charging response.
5. Dust accumulation in vents or casing.
6. Reduced storage space or app malfunction.

Benefits of Preventive Maintenance

- Keeps devices in good working condition.
- Reduces the frequency of faults and breakdowns.
- Saves time and cost of emergency repair.

- Enhances performance and efficiency.
- Ensures data safety and user satisfaction.
- Builds responsibility and professionalism in technicians.

Summary

Preventive maintenance involves **routine checks, cleaning, and updating** of computer and GSM devices to prevent faults.

It is cheaper and more effective than waiting for a system to fail.

Technicians and users should practice preventive maintenance regularly to extend device lifespan and ensure smooth operation.

Evaluation

1. Define preventive maintenance.
2. List four importance of preventive maintenance.
3. Mention two types of preventive maintenance.
4. State three preventive maintenance procedures for computers.
5. State three preventive maintenance procedures for GSM devices.

Assignment

1. Write out five safety precautions during preventive maintenance.
2. Mention five tools used for maintaining GSM and computer systems.
3. Explain three ways preventive maintenance helps to reduce repair costs.

WEEK 11: REVISION

Introduction

The revision week provides an opportunity for students to **review and consolidate** all they have learned during the third term in *Computer Hardware and GSM Repairs*.

It allows them to recall important facts, definitions, procedures, and safety precautions covered in both theory and practical lessons.

This final preparation will help students perform well in their end-of-term examinations and practical tests.

Summary of the Term's Topics

1. Computer Storage Devices

- **Definition:** Devices used to store data and information permanently or temporarily.
- **Examples:** Hard disk, Solid State Drive (SSD), Flash drive, and Memory card.
- **Functions:**
 - Hard disk stores operating system and software.
 - SSD provides faster access speed.
 - Flash drive transfers files between computers.
 - Memory card stores photos, music, and videos in GSM devices.

2. GSM Networks

- **Meaning:** GSM networks are communication systems that allow phones to send and receive voice, text, and data.
- **Types:**
 1. **2G (Second Generation):** Basic calls and text messages.
 2. **3G (Third Generation):** Supports video calls and mobile internet.

- 3. **4G (Fourth Generation):** Faster browsing and streaming.
- 4. **5G (Fifth Generation):** Ultra-fast internet and smart device connection.
- **Importance:** Enhances global communication, data transfer, and mobile services.

3. Assembling a Simple Computer

- **Meaning:** The process of connecting different computer components together to form a complete working system.
- **Major Components:**
 - System unit
 - Monitor
 - Keyboard
 - Mouse
 - Power supply unit
- **Steps:**
 1. Fix motherboard and CPU.
 2. Install RAM and power cables.
 3. Connect monitor, keyboard, and mouse.
 4. Power on and test functionality.
- **Safety Tips:**
 - Work on a clean, static-free surface.
 - Switch off power before connecting components.

4. Common GSM Problems

- **Screen Damage:** Broken or cracked screen due to fall or pressure.
- **Battery Fault:** Battery drains fast or fails to charge.
- **Charging Port Problem:** Poor or no connection during charging.
- **Speaker or Microphone Fault:** Affects sound input or output.
- **Solutions:**
 - Replace damaged parts.
 - Clean charging ports.
 - Use compatible batteries and accessories.

5. Practical – Replacing GSM Parts

- **Parts Commonly Replaced:** Battery, screen, keypad, charging port, and speakers.
- **Tools Required:** Screwdriver set, pry tools, tweezers, suction cup, and soldering iron.
- **Procedure:**
 1. Switch off phone.
 2. Disassemble carefully.
 3. Remove and replace faulty part.
 4. Reassemble and test the phone.
- **Safety Measures:** Avoid force, use correct tools, handle screen gently.

6. Introduction to Soldering

- **Meaning:** Soldering is the process of joining two or more metal parts by melting solder around them.
- **Tools:** Soldering iron, solder wire, flux, desoldering pump, sponge.

- **Safety:** Avoid burns, ensure ventilation, and switch off when not in use.
- **Uses in GSM Repair:** Fixing loose connections on the circuit board.

7. Practical – Soldering Simple Joints

- **Types of Joints:**

- Lap joint
- T-joint
- Twist joint
- Butt joint

- **Steps:**

1. Strip wires and clean surfaces.
2. Apply flux and heat with soldering iron.
3. Apply solder and allow to cool.
4. Inspect for a smooth, shiny joint.

- **Common Faults:** Cold joint, short circuit, excess solder.

- **Correction:** Heat properly, use right amount of solder, and avoid bridging.

8. Preventive Maintenance

- **Meaning:** Regular care to prevent faults before they occur.

- **For Computers:**

- Clean hardware regularly.
- Update antivirus and system software.
- Backup data.

- **For GSM Phones:**
 - Avoid overcharging.
 - Keep away from water and dust.
 - Use protective cases.
- **Benefits:** Saves cost, increases lifespan, improves performance.

General Safety Precautions in Computer and GSM Repair

1. Switch off and unplug devices before working on them.
2. Use the correct tools for each operation.
3. Avoid direct contact with live circuits.
4. Work on a clean, dry surface.
5. Keep small parts and screws safely in containers.
6. Handle circuit boards and screens with care.
7. Avoid overheating components when soldering.
8. Follow standard repair procedures.